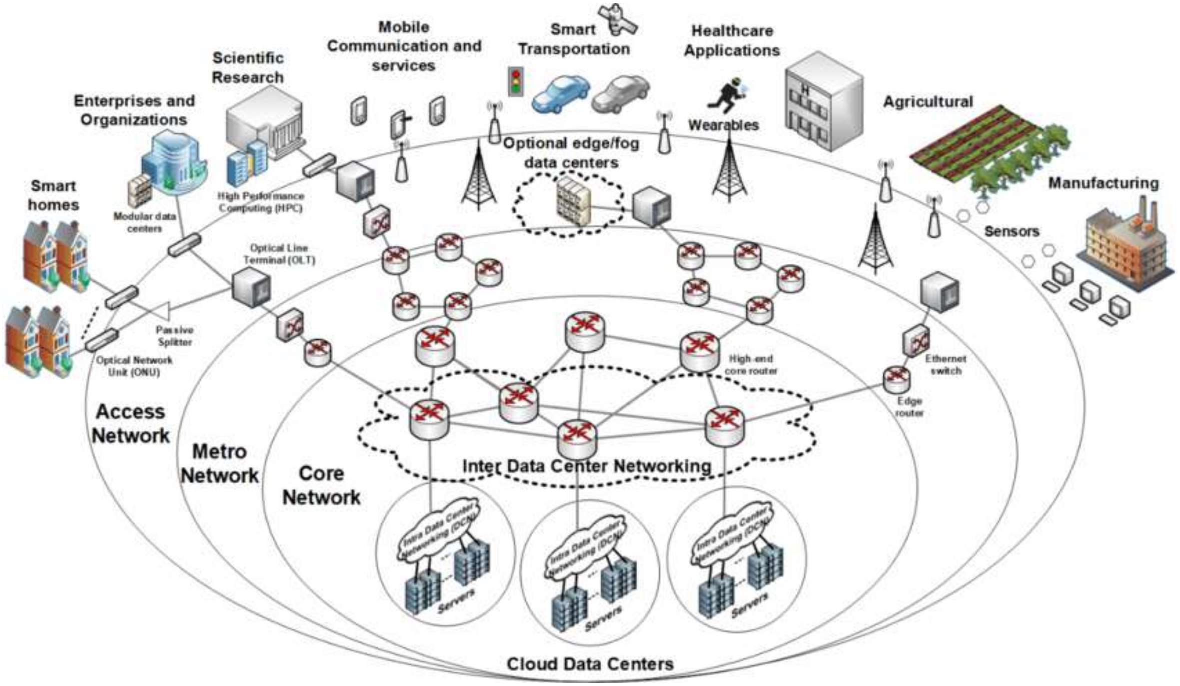
Data Communications and Networking

Chapter 1.1: Basic Concepts

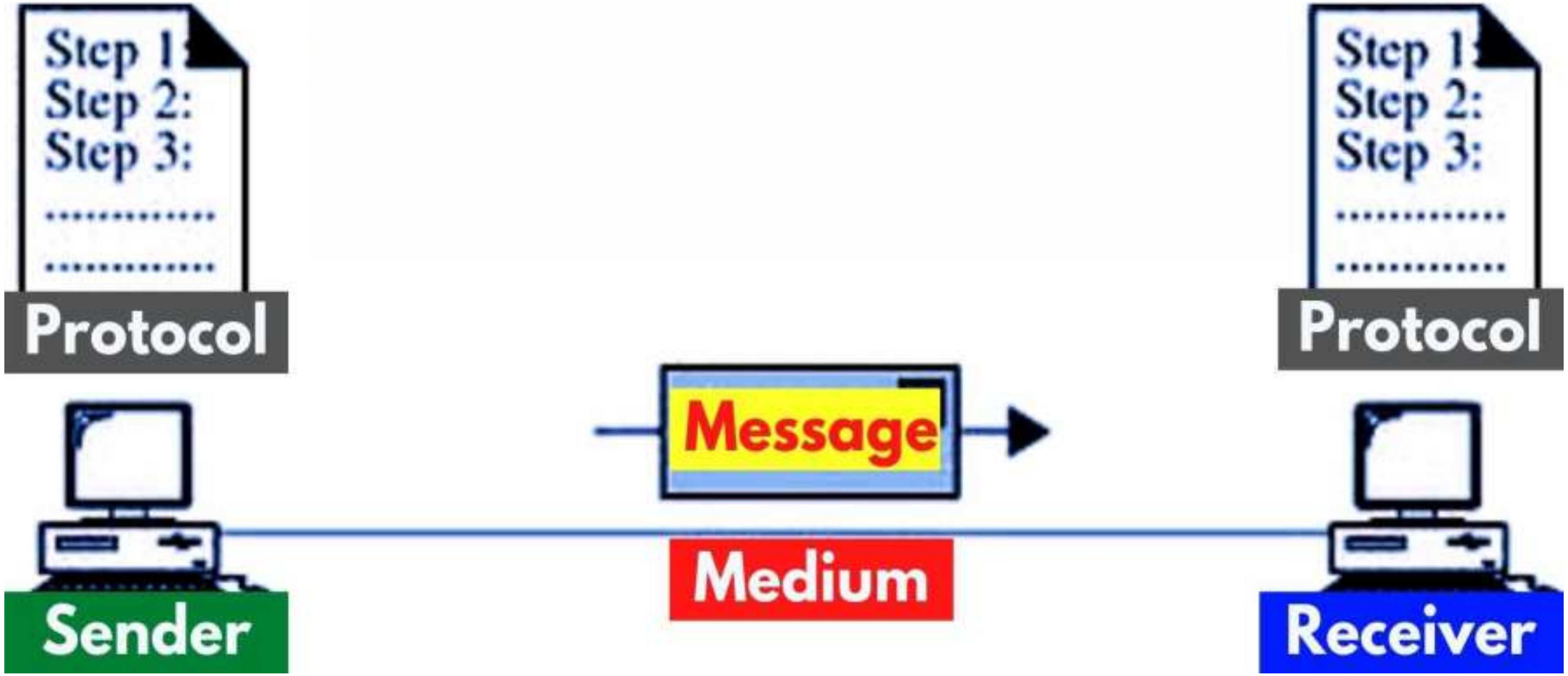
Data Communication

☐ Definition

- Data communication is the process of transferring data from one place to another or between two locations.
- It allows electronic and digital data to move between two networks, no matter where the two are located geographically, what the data contains, or what format they are in.



Five Components of Data Communication

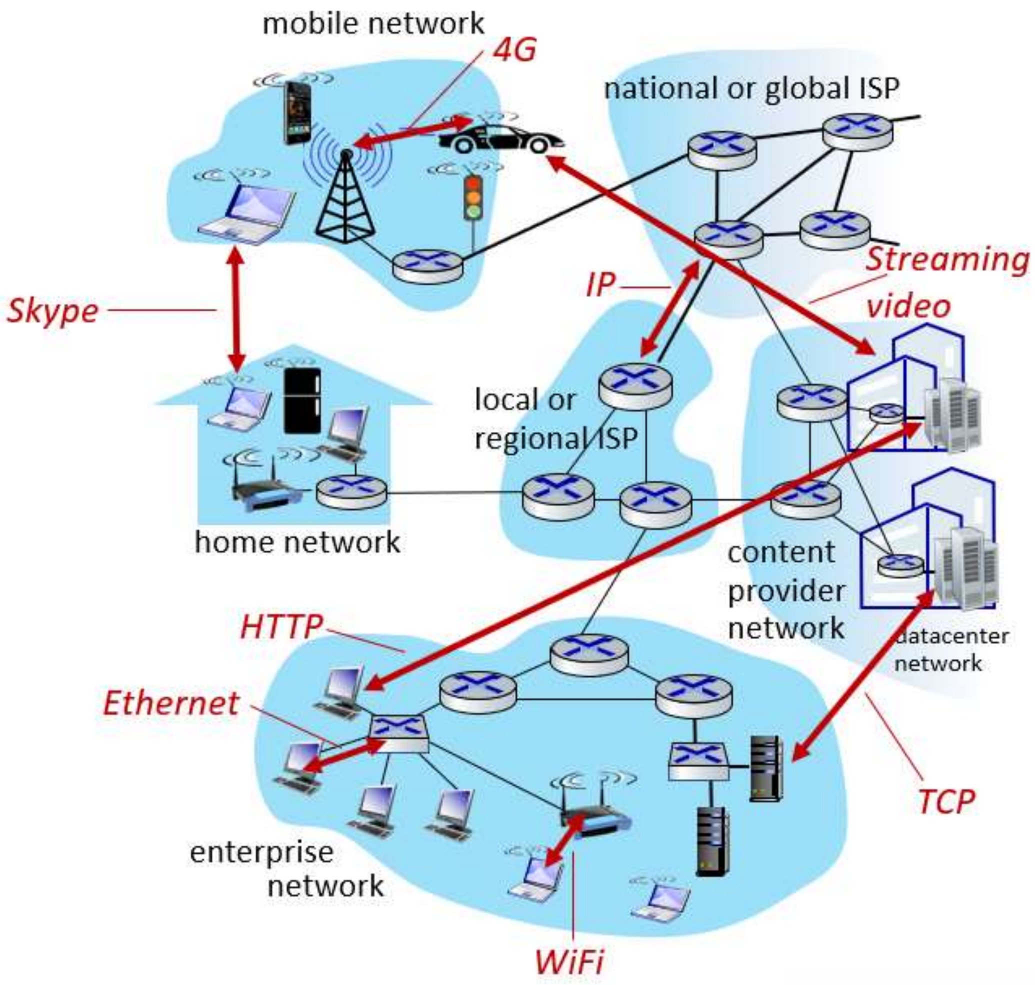


Q: Type of data format? Text?

What is Protocol?

☐ Definition

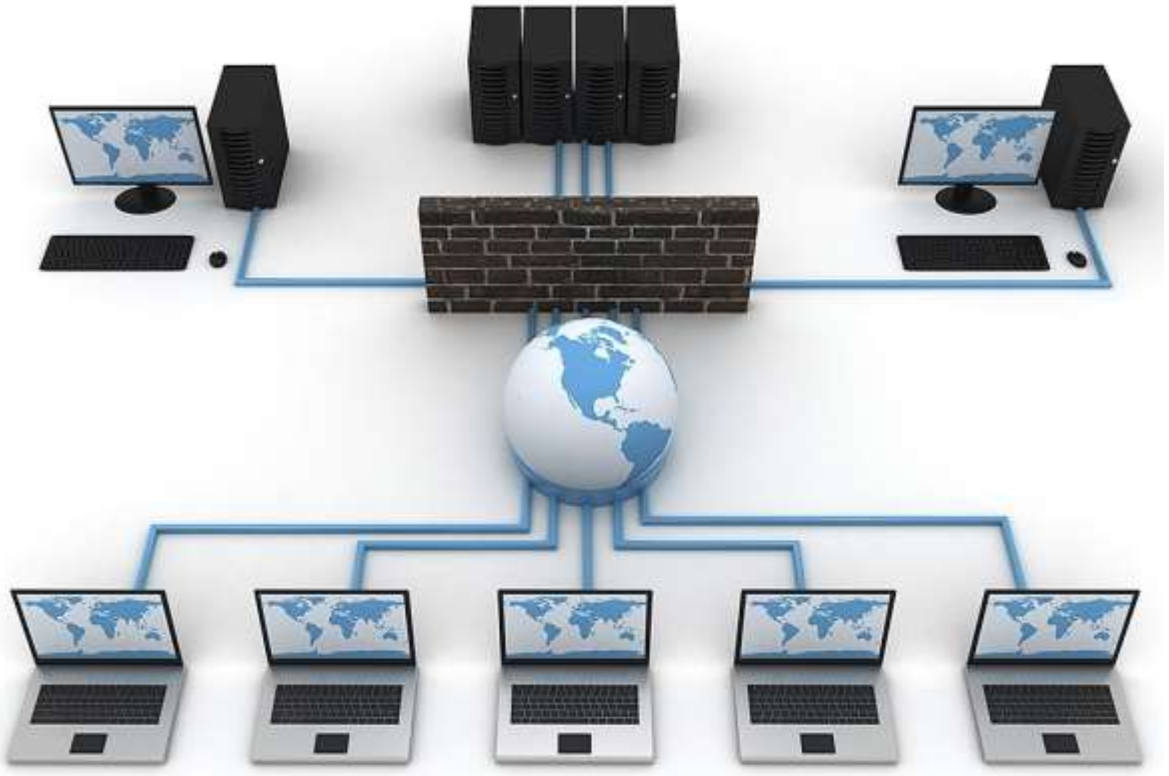
- Protocols define the format, order of messages sent and received among network entities, and actions taken on message transmission, receipt
- It makes data communication between different nodes, e.g., computers, servers, routers and virtual machines possible
- It includes rules of data sequencing, data routing, flow control, error correction, etc



Computer Network

☐Definition

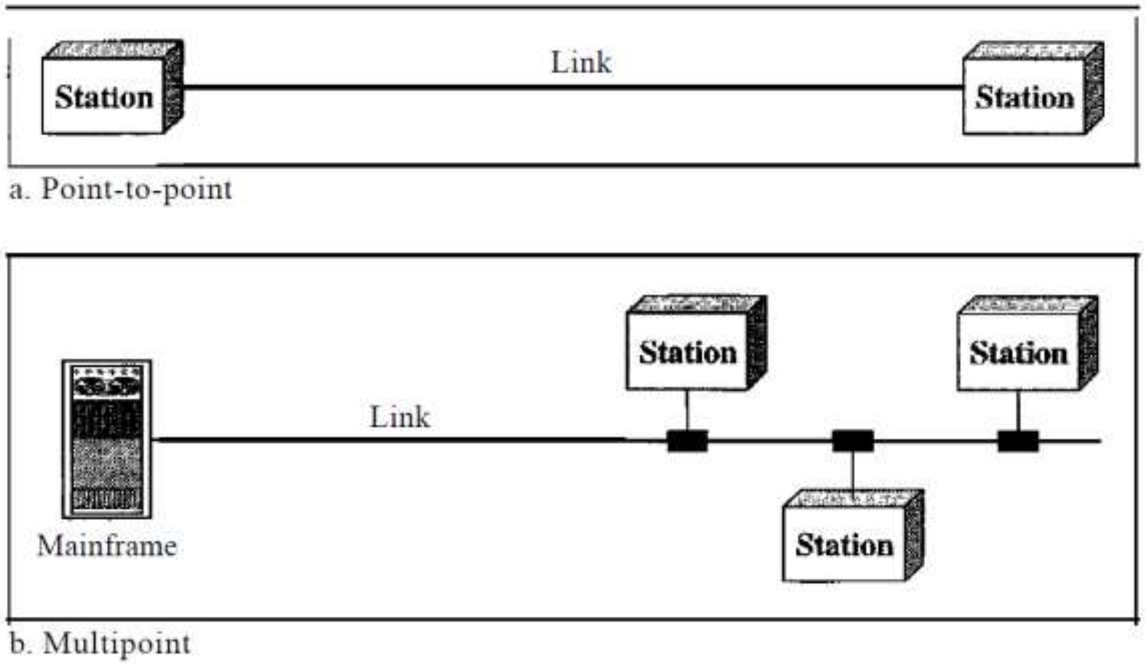
- A **computer network** is a set of computers sharing resources located on or provided by network nodes.
- Nodes use common communication protocols over digital interconnections to communicate with each other
- Telecommunication network technologies based on physically wired, optical, and wireless radio-frequency methods that may be arranged in a variety of network topologies



Physical Structures

☐Type of Connection

- **Point-to-Point**: a dedicated link between two devices, can be connected by wire or cable, also microwave/satellite
- **Multipoint**: more than two specific devices share a single link. the capacity of the channel is shared, either spatially or temporally



Categories of Topology

Bus Topology

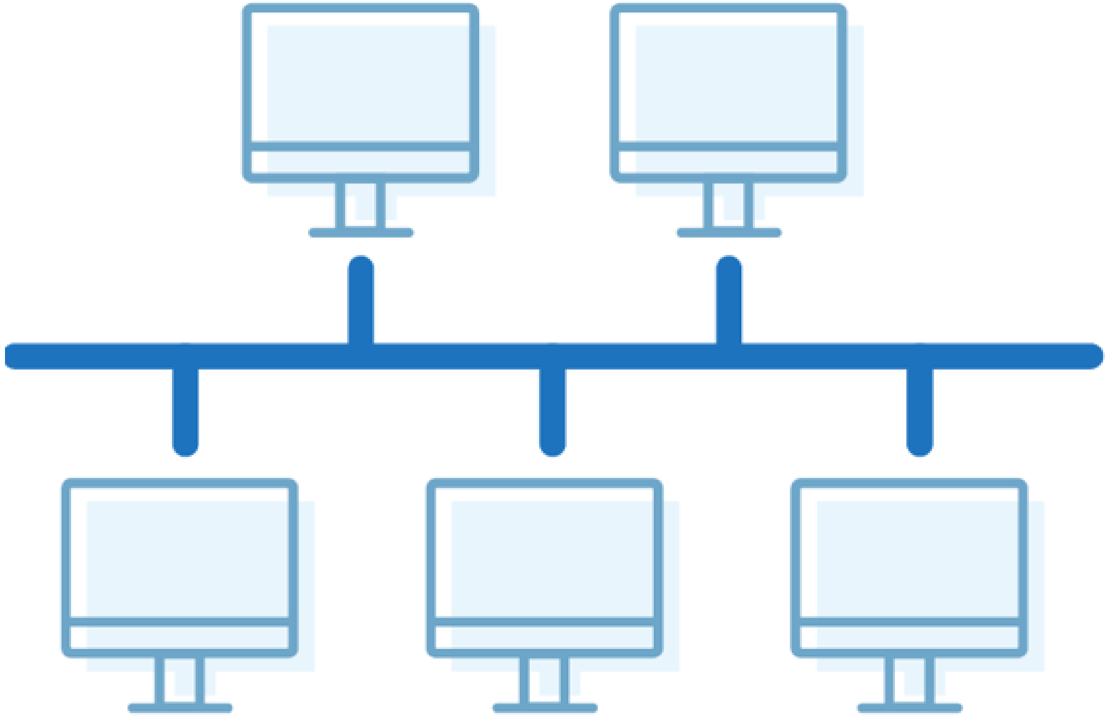
- All devices are connected to a common backbone

Pros

- Less cable length
- Easy to connect

Cons

- Sensitive to the state of main backbone/cable
- Difficult to identify problem



Categories of Topology ★

Ring Topology

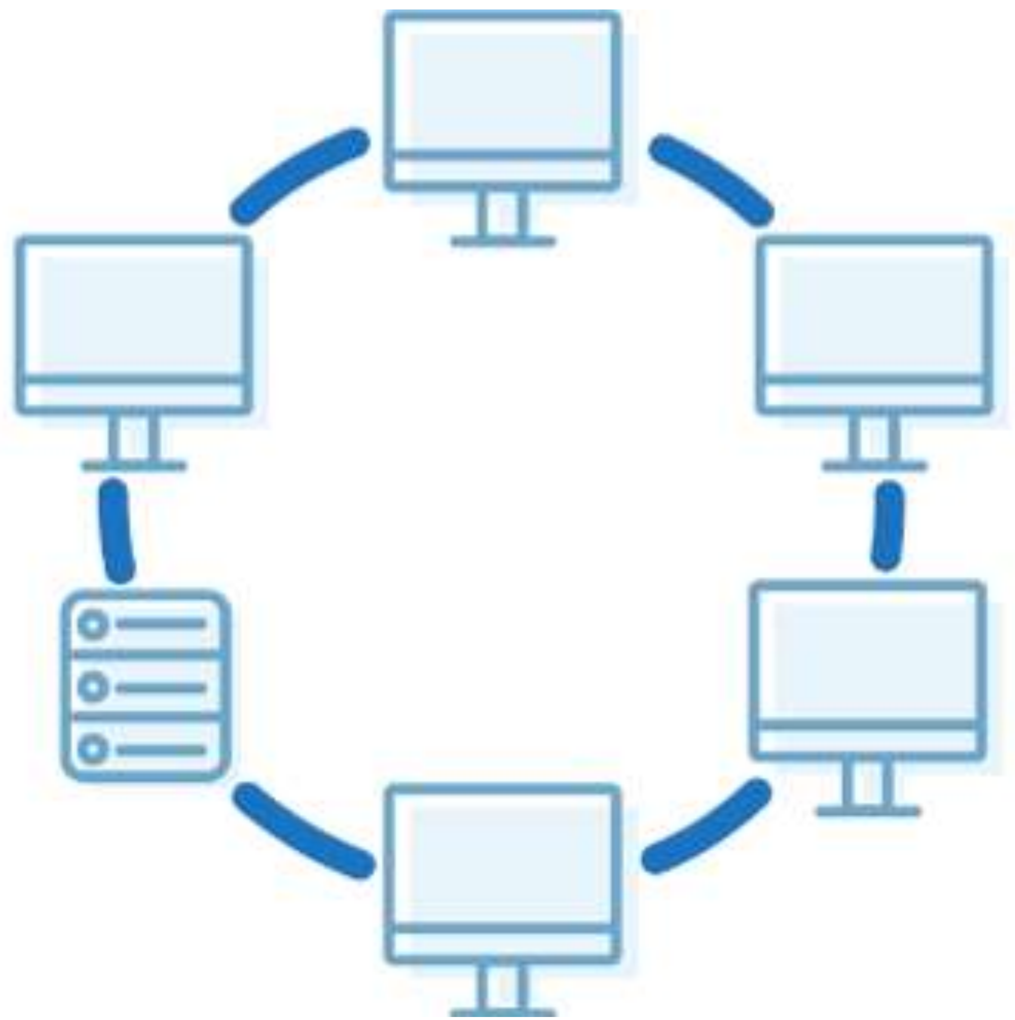
- Every device has exactly two neighbors, messages travel through a ring in the same direction

Pros

- Easy to install and reconfigure
- Fault isolation is simplified

Cons

- Unidirectional traffic



Categories of Topology

Star Topology

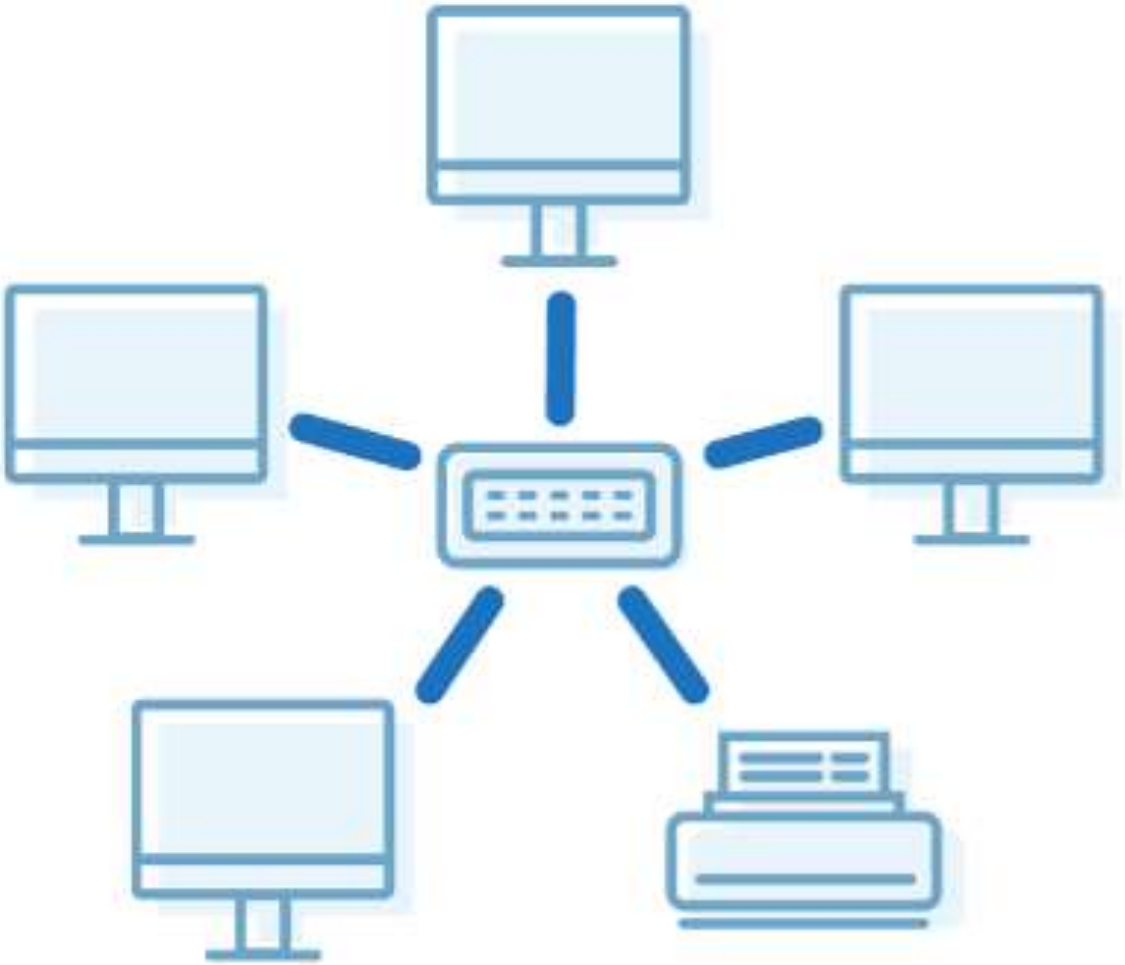
- Each node (file server, workstations, and peripherals) is connected directly to a central network hub or concentrator

Pros

- Robustness
- Easy fault identification

Cons

- Dependency of the whole topology on one single point, the hub



Categories of Topology

Tree Topology

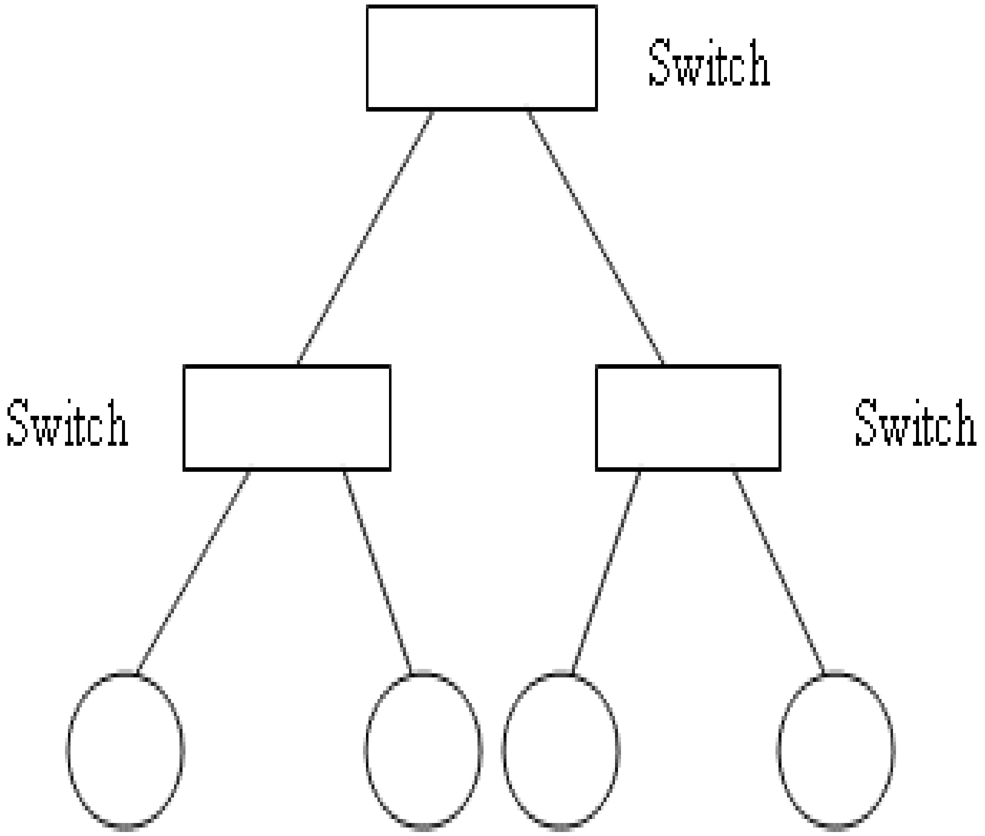
- Tree topologies allow for the expansion of an existing network. For example, enabling schools to configure a network for labs.

Pros

- Point-to-point wiring for individual segments

Cons

- If the backbone line breaks, the entire segment goes down.



Categories of Topology ★

❑Mesh Topology

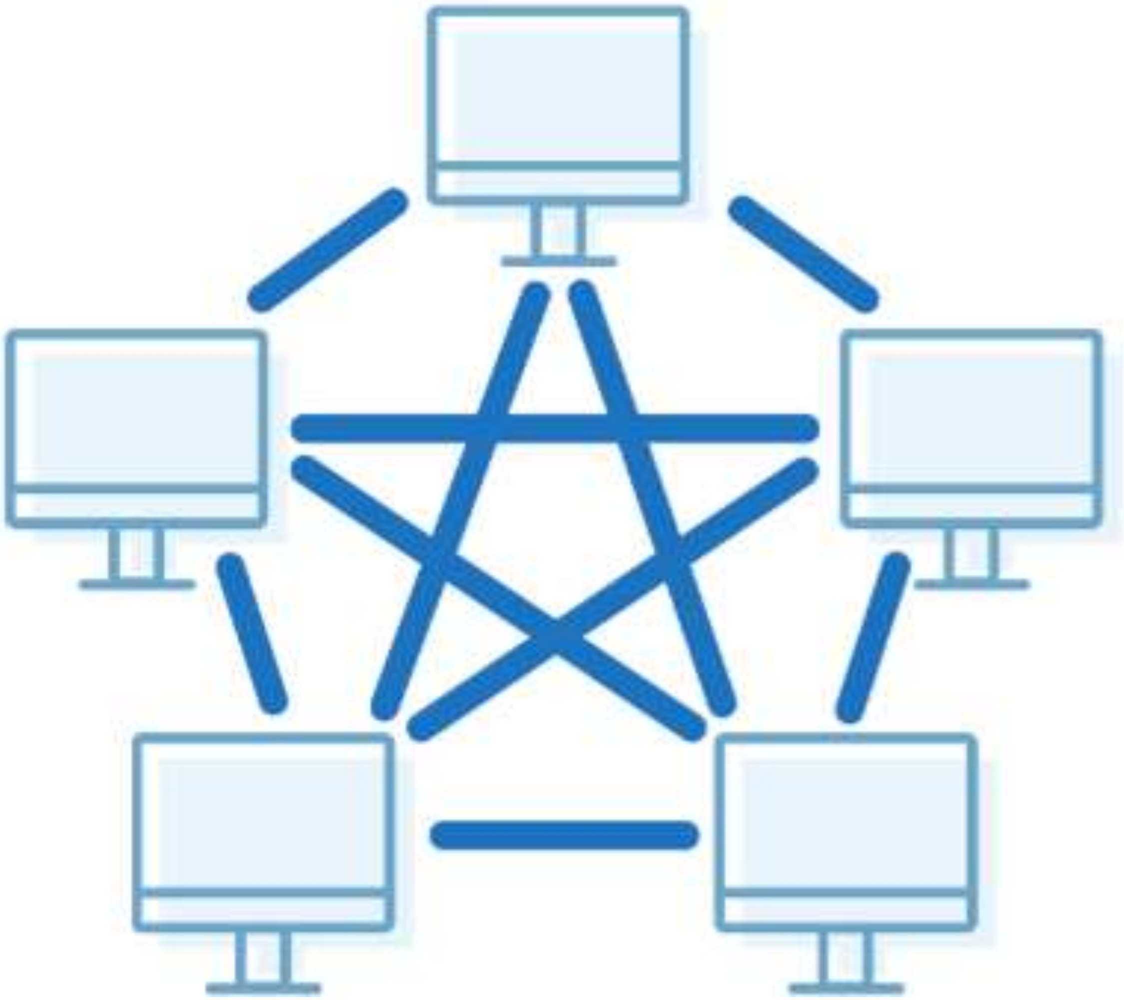
- Dedicated point-to-point link to every other device

❑Pros

- Each connection carries its own data load
- Easy fault identification and isolation
- Robust

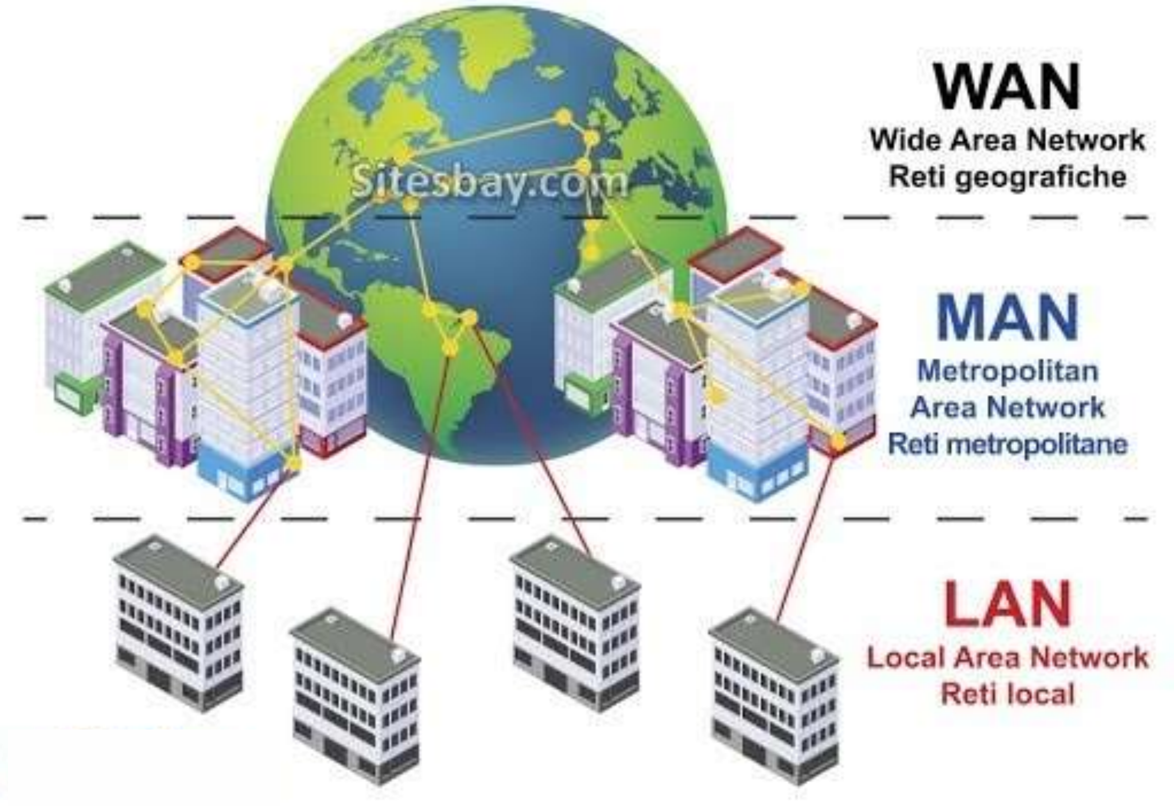
❑Cons

- Installation and reconnection are difficult
- Expensive hardware, cable, wall, etc



Categories of Network

- Local Loop
- Local Area Network
- Metropolitan Area Network
- Wide Area Network



Local Loop

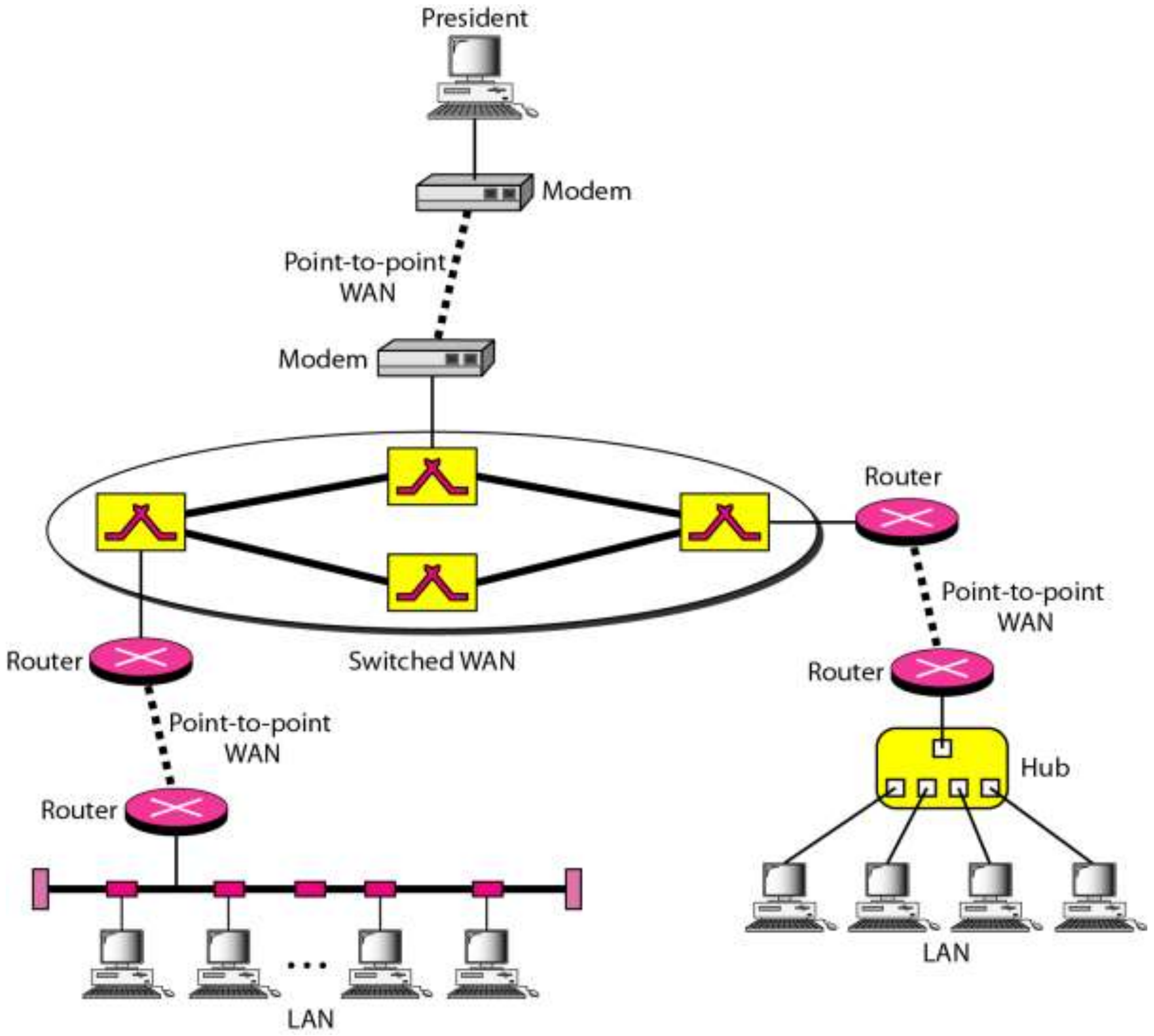
- it refers to the “last mile” of analog phone line that goes from the Exchange Centre to your house.
  - Voice lines
  - Modem (56 kbps)
  - ISDN (Integrated Services Digital Network): 2 x 64 kbps digital lines
  - ADSL (Asymmetrical Digital Subscriber Line): up to 8 Mbps
  - Cable Modems (up to 30 Mbps)
  - Network cable

Categories of Network

Comparison	LAN	MAN	WAN
Full Name	Local Area Network	Metropolitan Area Network	Wide Area Network
Meaning	A network that connects a group of computers in a small geographical area	It covers relatively large region such as cities, towns	It spans large locality & connects countries together. e.g. Internet
Ownership of Network	Private	Private or Public	Private or Public (VPN)
Design and Maintenance	Easy	Difficult	Difficult
Propagation Delay	Short	Moderate	Long
Speed	High	Moderate	Low
Equipment Used	NIC, Switch, Hub	Modem, Router	Microwave, Radio Transmitter & Receiver
Range(Approximately)	1 to 10 km	10 to 100 km	Beyond 100 km
Used for	College, School, Hospital	Small towns, City	State, Country, Continent

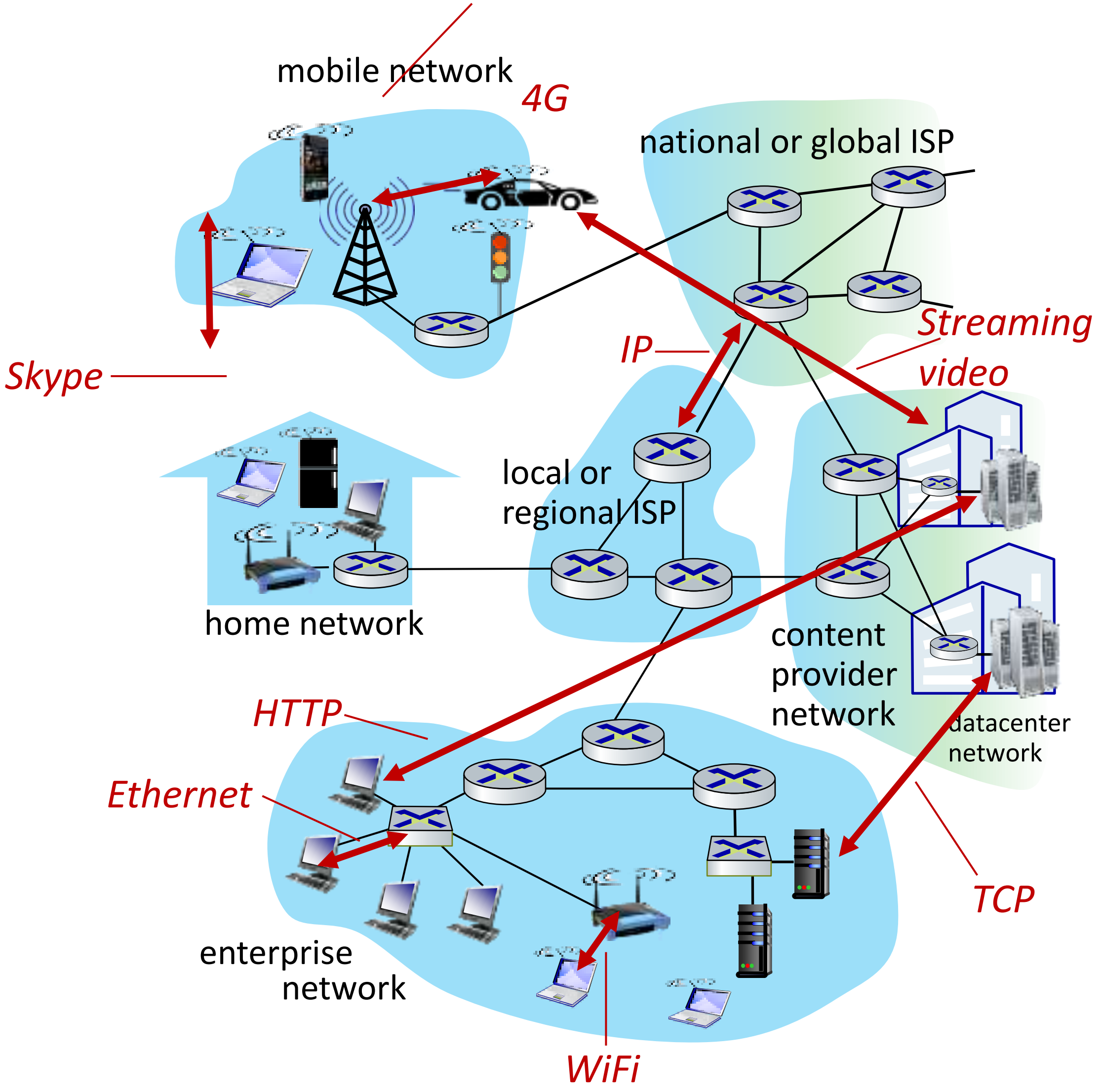
Categories of Network

A heterogeneous network made of four WANs and two LANs



The Internet: a "nuts and bolts" view

- **Internet: "network of networks"**
  - Interconnected ISPs
- **protocols are everywhere**
  - control sending, receiving of messages
  - e.g., HTTP (Web), streaming video, Skype, TCP, IP, WiFi, 4/5G, Ethernet



Why layering?

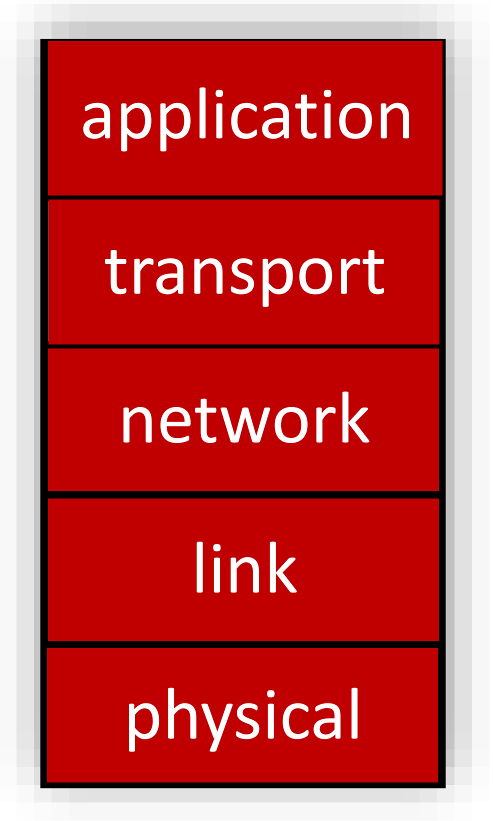
Approach to designing/discussing complex systems:

- **explicit structure allows identification, relationship of system's pieces**
  - layered reference model for discussion

- **modularization eases maintenance, updating of system**
  - change in layer's service implementation: transparent to rest of system
  - e.g., change in gate procedure doesn't affect rest of system

Layered Internet Protocol Stack

- *application:* supporting network applications
  - HTTP, IMAP, SMTP, DNS
- *transport:* process-process data transfer
  - TCP, UDP
- *network:* routing of datagrams from source to destination
  - IP, routing protocols
- *link:* data transfer between neighboring network elements
  - Ethernet, 802.11 (WiFi), PPP
- *physical:* bits “on the wire”



Transportation Services

◻ Connection-oriented services:

- To establish a connection, the sender, receiver and the subnet conduct a negotiation about parameters to be used, such as maximum message size, quality of service required, etc.
- The **Quality of Service** can be characterized in terms of Data Loss Rate, Delay Jitter, Data Rate, etc.
- Application Example: File transfer

◻ Connectionless service

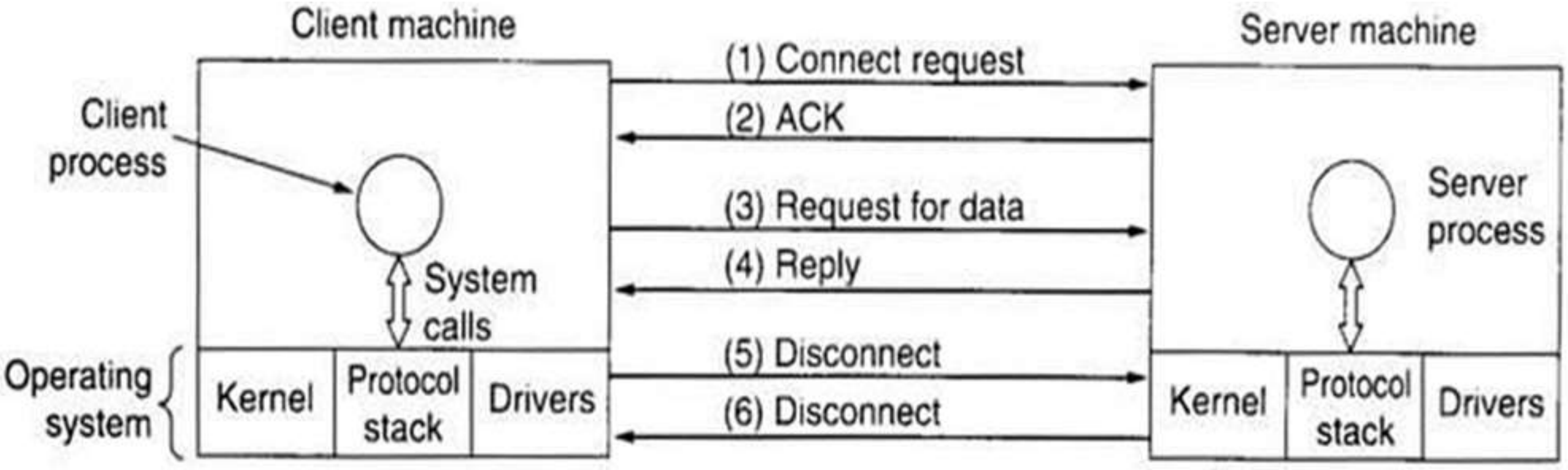
- Each message carries the full destination address, and each one is routed through the system independent of all the others. No negotiation is needed between sender and receiver.

Service Primitives

- A service is formally specified by a set of primitives (operations) available to a user process to access the service. These primitives tell the service to perform some action or report on an action taken by a peer entity.
- Example: service primitives that might be provided to implemented a reliable byte stream in a client-server environment:

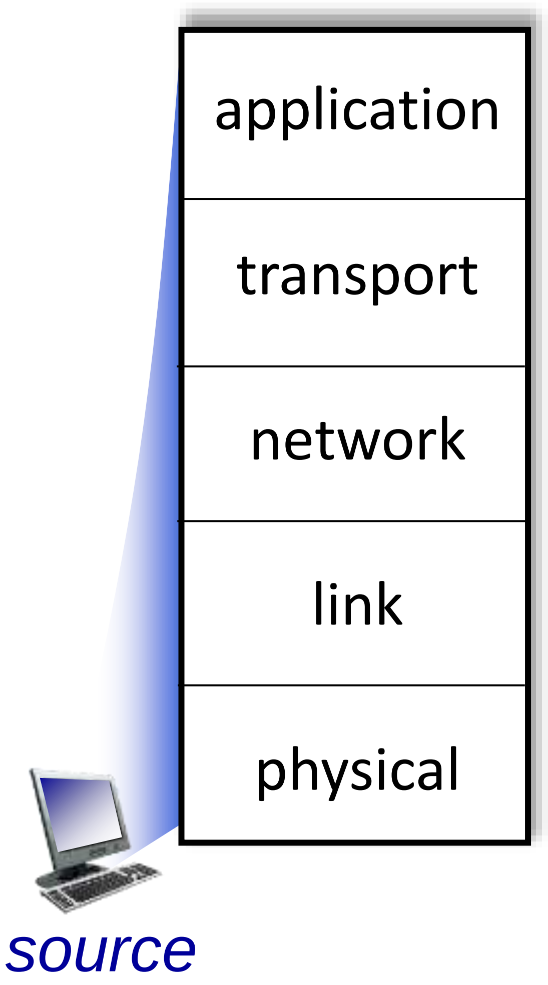
Primitives	**Meaning**
LISTEN	Block waiting for an incoming connection
CONNECT	Establish a connection with a waiting peer
RECEIVE	Block waiting for an incoming message
SEND	Send a message to the peer
DISCONNECT	Terminate a connection

Service Primitives



Example: Client-server interactions

Services, Layering and Encapsulation



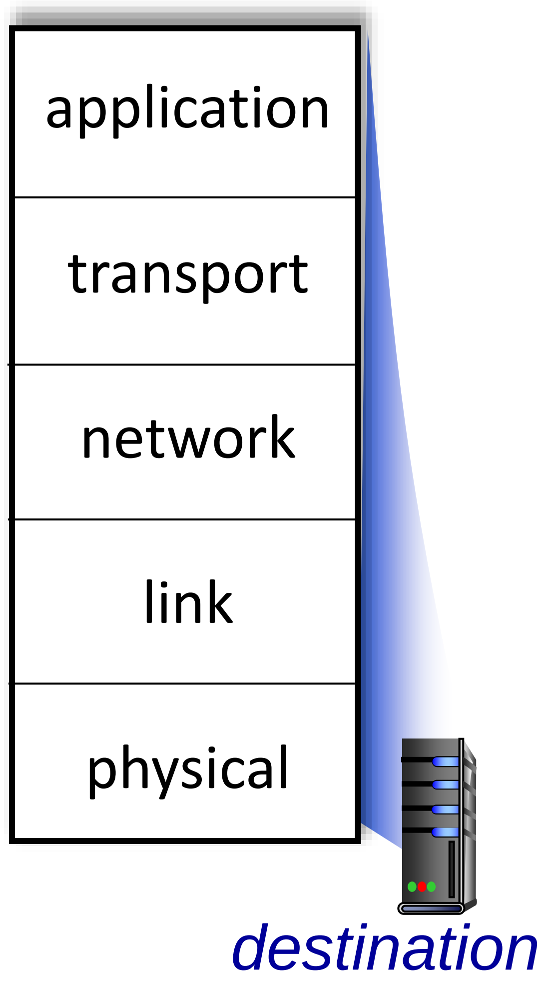
M

Application exchanges messages to implement some application service using *services* of transport layer

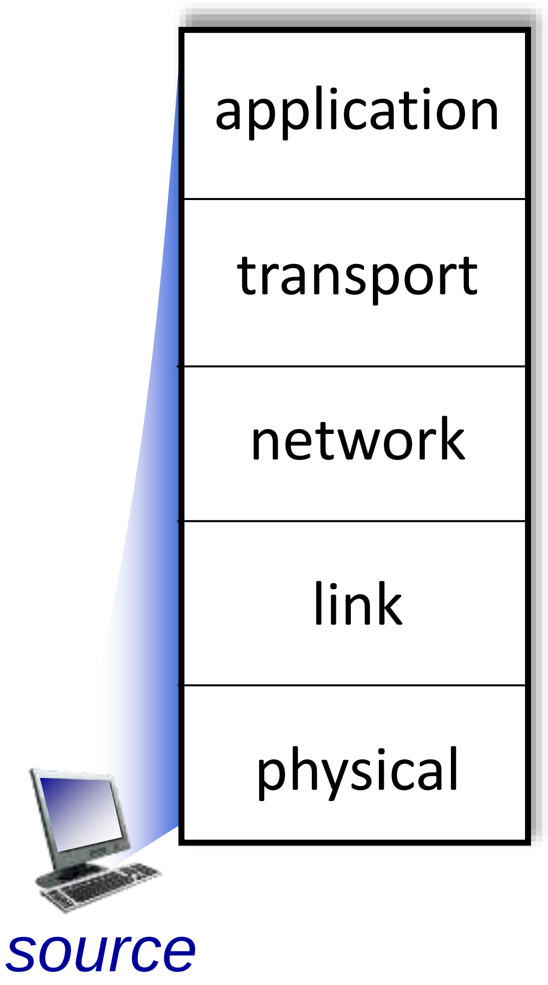
H_t M

Transport-layer protocol transfers M (e.g., reliably) from one *process* to another, using services of network layer

- transport-layer protocol encapsulates application-layer message, M, with *transport* layer-layer header H_t to create a transport-layer segment
  - H_t used by transport layer protocol to implement its service



Services, Layering and Encapsulation



M

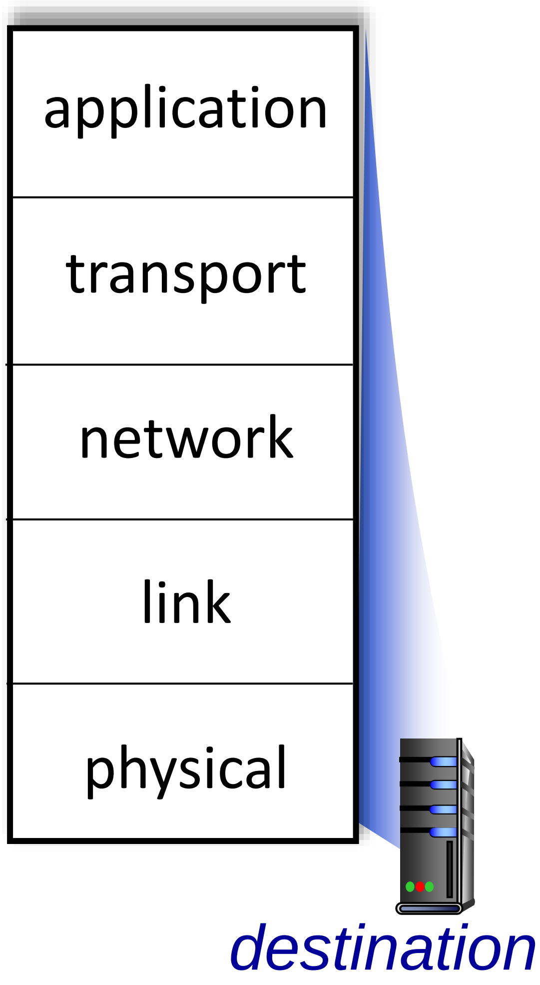
H_t M

Transport-layer protocol transfers M (e.g., reliably) from one *process* to another, using services of network layer

H_n H_t M

Network-layer protocol transfers transport-layer segment [H_t | M] from one *host* to another, using link layer services

- network-layer protocol encapsulates transport-layer segment [H_t | M] with network layer-layer header H_n to create a network-layer datagram
  - H_n used by network layer protocol to implement its service



Services, Layering and Encapsulation

application | transport | network | link | physical

source

M

H_t M

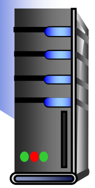
Transport-layer protocol transfers M (e.g., reliably) from one *process* to another, using services of network layer

H_n H_t M

Network-layer protocol transfers transport-layer segment [H_t | M] from one *host* to another, using link layer services

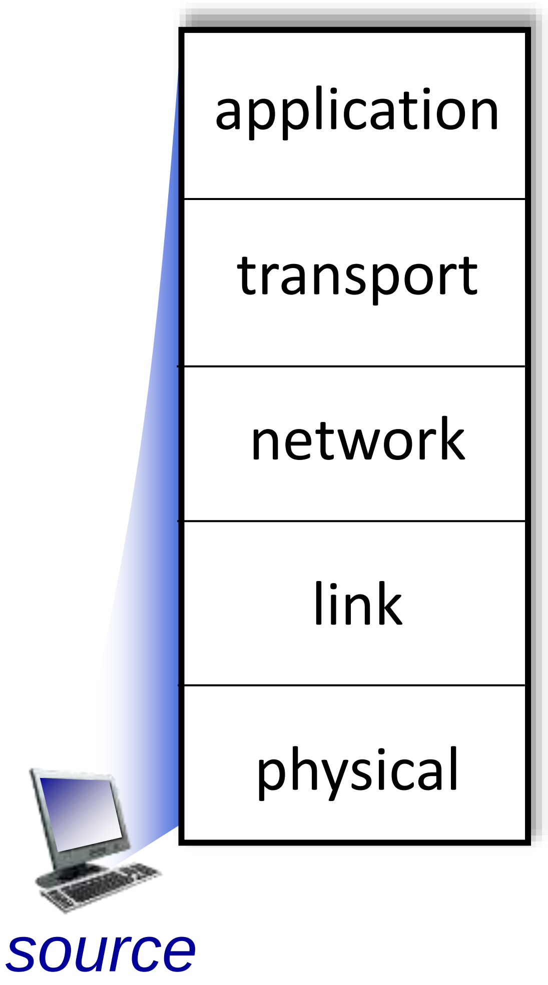
- network-layer protocol **encapsulates** transport-layer segment [H_t | M] with network layer-layer header H_n to create a network-layer datagram
  - H_n used by network layer protocol to implement its service

application | transport | network | link | physical



destination

Services, Layering and Encapsulation



M

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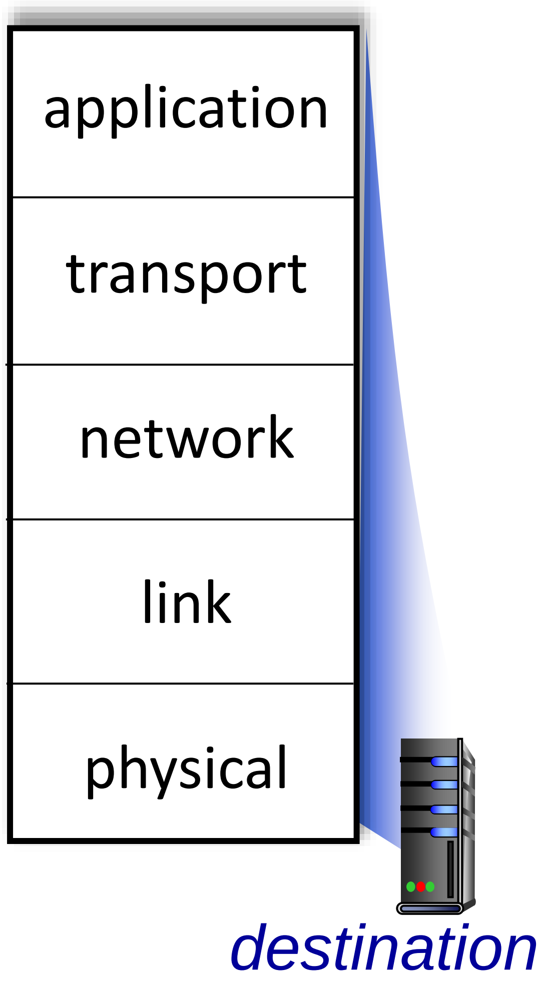
Transport-layer protocol transfers M (e.g., reliably) from one *process* to another, using services of network layer

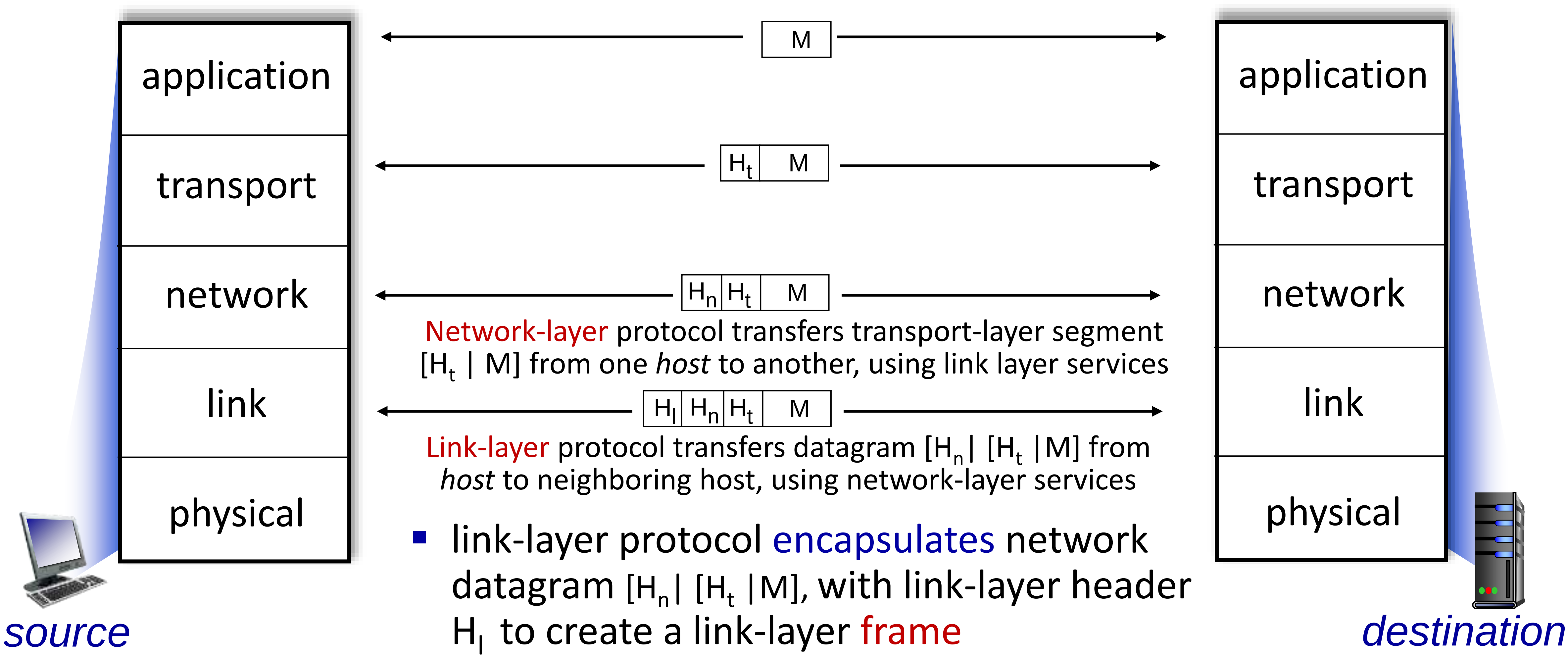
H_n H_t M

Network-layer protocol transfers transport-layer segment [H_t | M] from one *host* to another, using link layer services

- network-layer protocol encapsulates transport-layer segment [H_t | M] with network layer-layer header H_n to create a network-layer datagram
  - H_n used by network layer protocol to implement its service

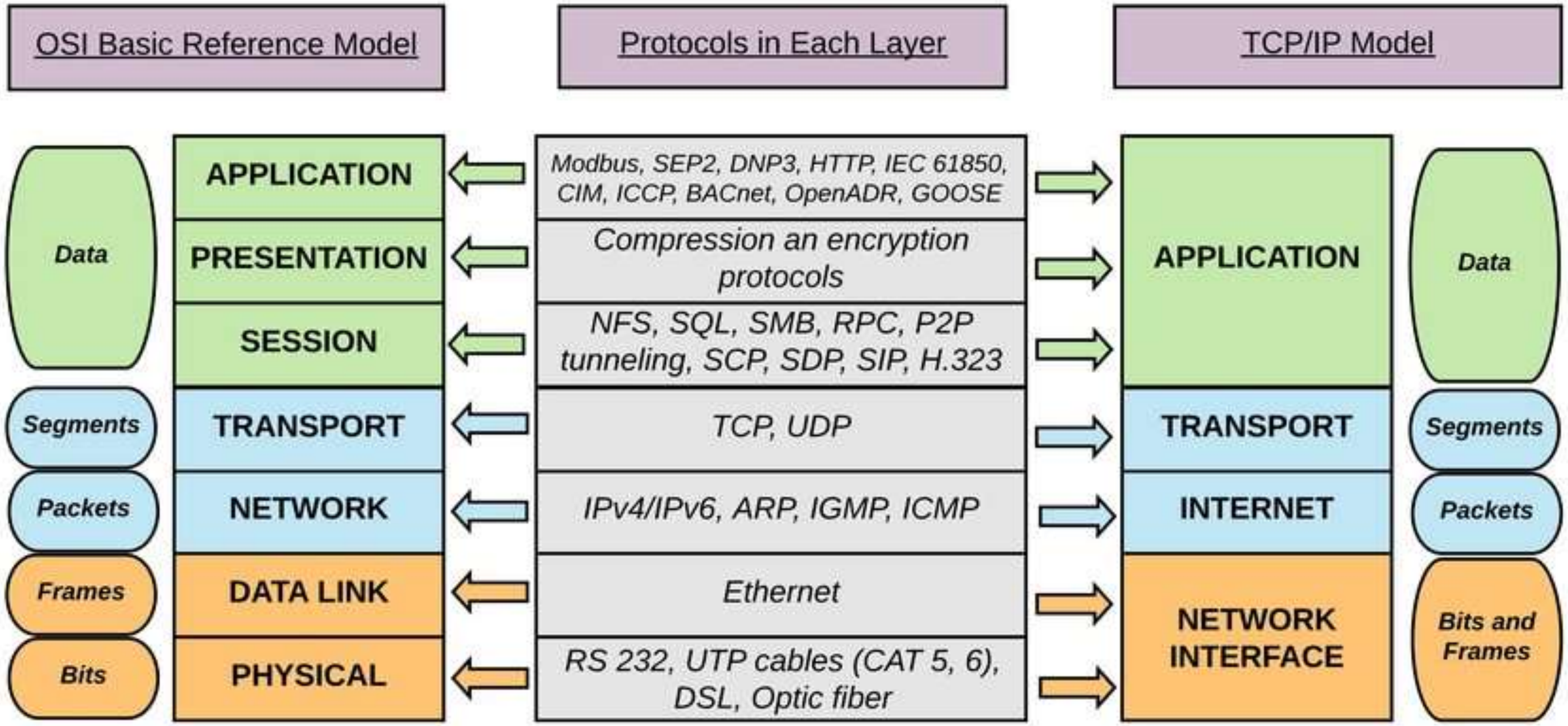


Services, Layering and Encapsulation



- link-layer protocol encapsulates network datagram [H_n| [H_t |M], with link-layer header H_l to create a link-layer frame

Two Reference Models



Network Interface = Data Link + Physical (TCP/IP overall 5 layers for teaching purpose)

Two Reference Models

- Open System Interconnection (OSI) Model has seven layers.
  - **Physical Layer:** It is concerned with transmitting raw bits over a communication channel. This layer deals with the issues including mechanical, electrical, timing interfaces, and the physical transmission medium.
  - **Data Link Layer:** It is to transform a raw transmission facility into a line that appears free of undetected transmission errors to the network layer.
  - **Network Layer:** It controls the operation of the subnet. It routes packets from source to destination. The routing algorithm can be static or dynamic.

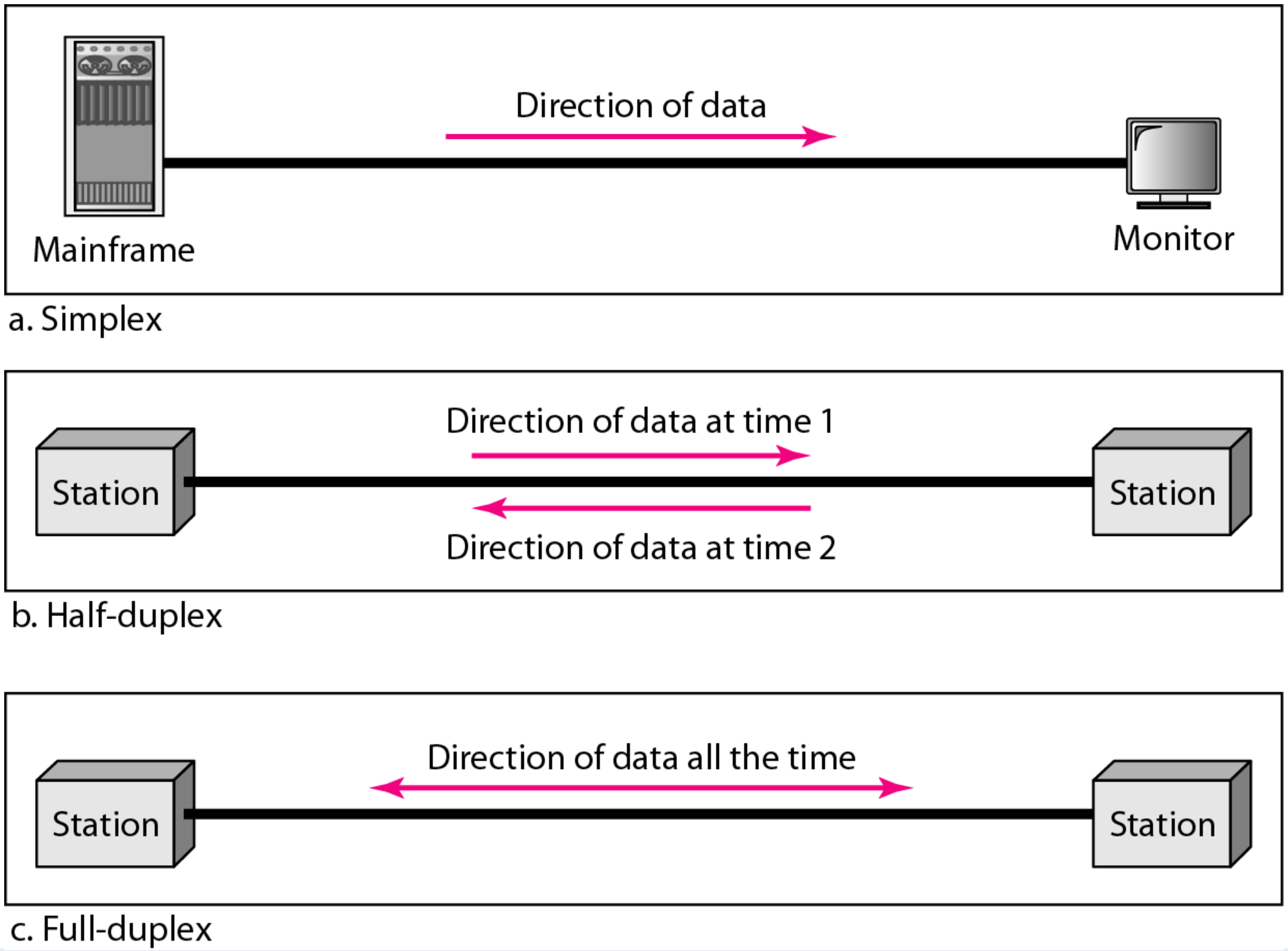
Two Reference Models

- **Transport Layer:** It is to accept data from above, split it up into smaller units if need be, pass these to the network layer, and ensure that the pieces all arrive correctly at the other hand.
- **Session Layer:** It allows users on different machines to establish sessions between them.
- **Presentation Layer:** It is concerned with the syntax and semantics of the information transmitted.
- **Application Layer:** It contains a variety of protocols that are commonly needed by users, such as HTTP, Email, etc.

Data Communications and Networking

Chapter 1.2: Data and Signals

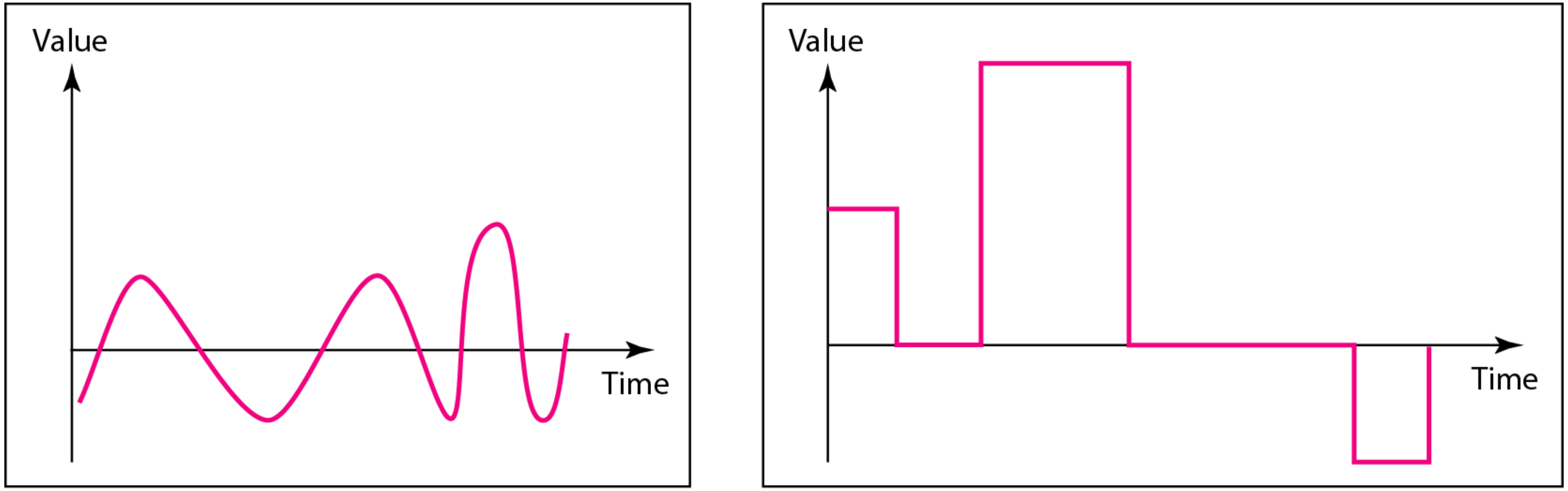
Data Flow



Direction of data
Mainframe
Monitor
a. Simplex
Direction of data at time 1
Station
Station
Direction of data at time 2
b. Half-duplex
Direction of data all the time
Station
Station
c. Full-duplex

Data and Signals

- **To be transmitted, data must be transformed to electromagnetic signals**
- **Data and Signals can be analog or digital**
  - **Analog** refers to information that is continuous
  - **Digital** refers to information that has discrete states



a. Analog signal   b. Digital signal

Analog vs Digital

Analog

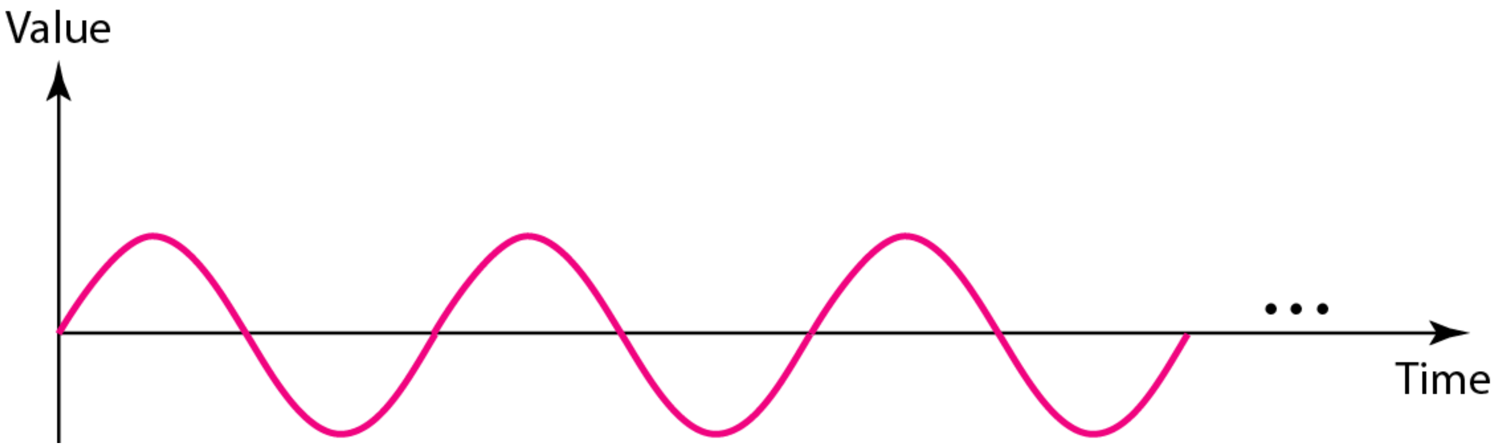
- Natural Representation: directly represent continuous variations in real-world phenomena (e.g., sound, light)
- Simple Processing: circuits are often less costly
- Efficient Bandwidth Utilization: In some cases, analog signals can be more bandwidth-efficient, especially in high-frequency applications

Digital

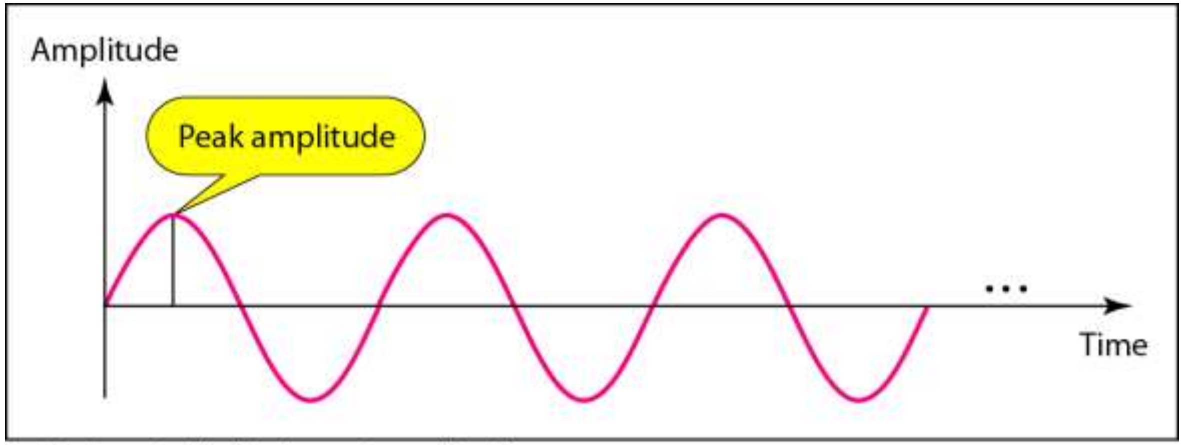
- Noise Resistance: much more resistant to noise and interference
- Easy to Store and Copy: easily stored, transmitted, and
- Flexibility: can be processed through software
- Compression and Encryption: can be compressed to save space and encrypted for security

Periodic Analog Signals

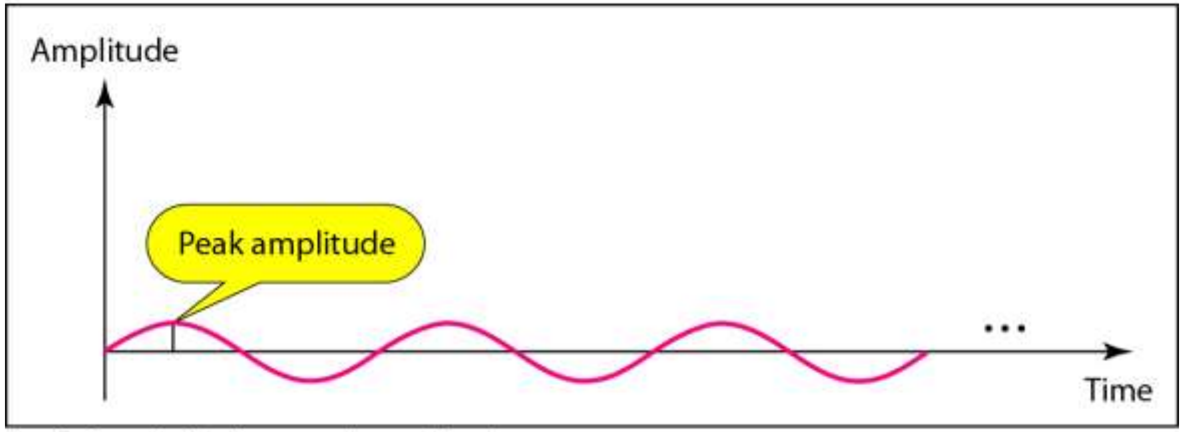
- **In data communications, we commonly use periodic analog signals and nonperiodic digital signals**
  - A simple periodic analog signal, a sine wave, cannot be decomposed into simpler signals



Signal Amplitude



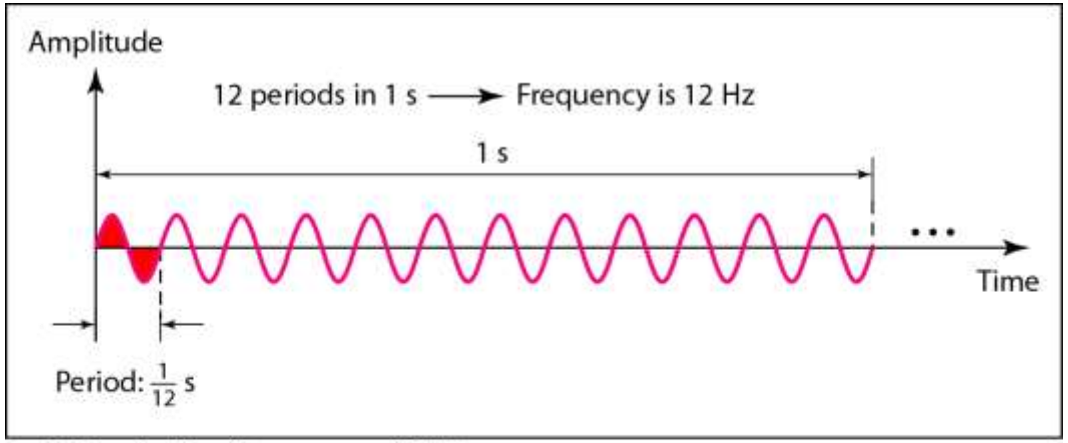
a. A signal with high peak amplitude



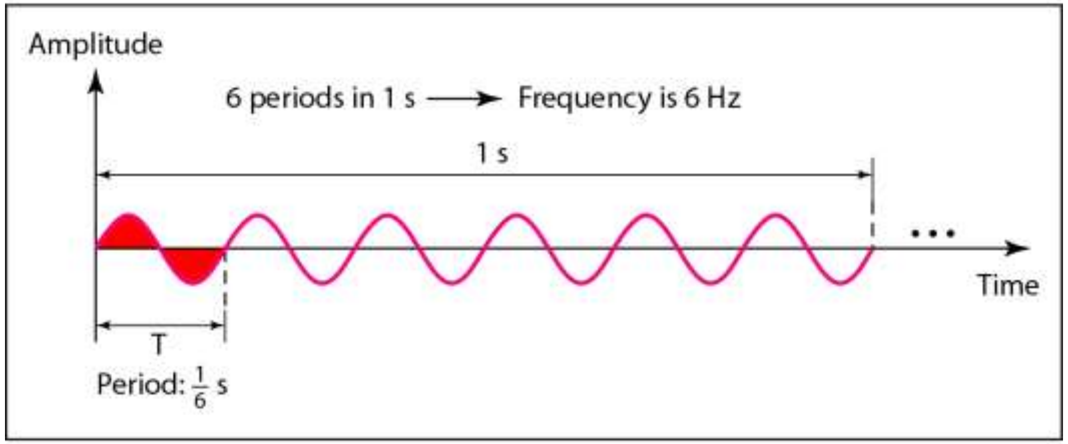
b. A signal with low peak amplitude

Two signals with different amplitudes

Signal Frequency



a. A signal with a frequency of 12 Hz



b. A signal with a frequency of 6 Hz

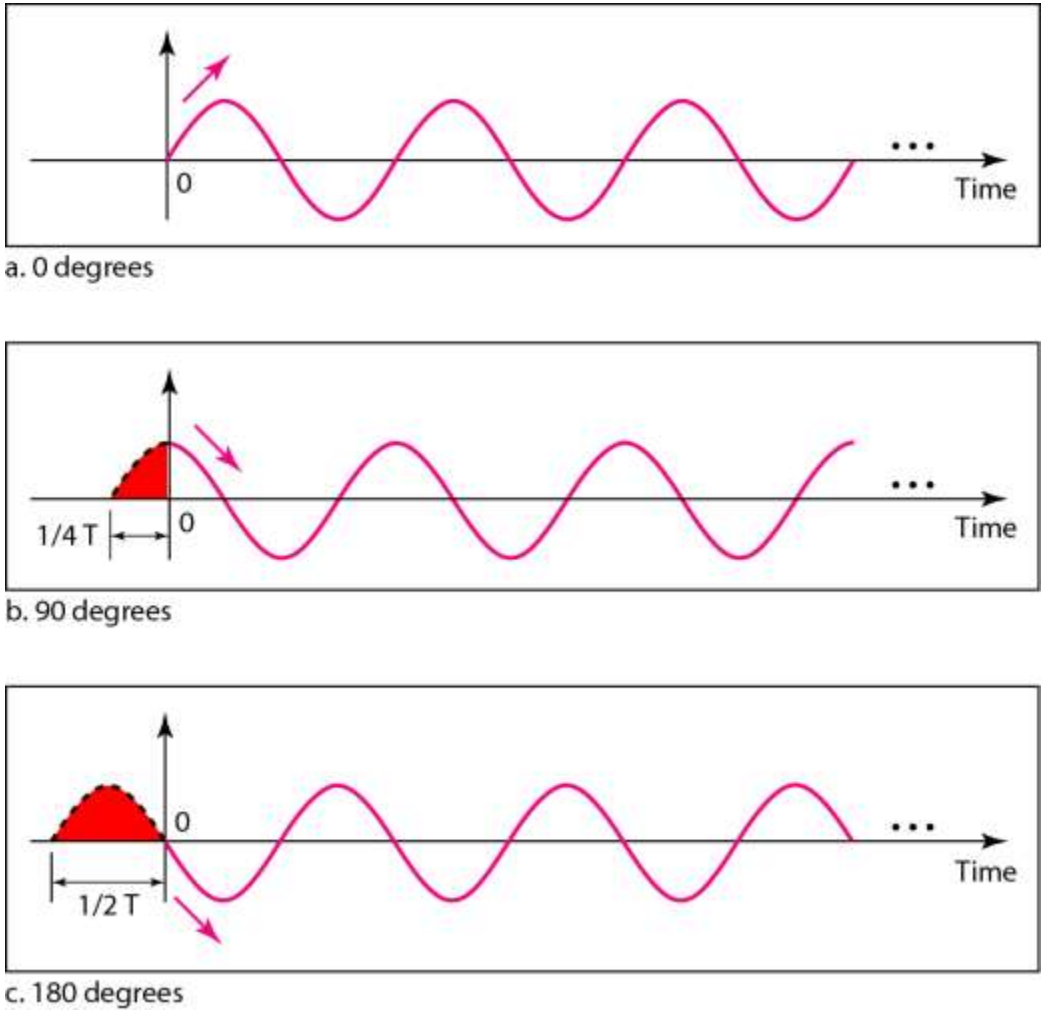
Two signals with different frequencies

Signal Frequency

Unit	*Equivalent*	*Unit*	*Equivalent*
Seconds (s)	1 s	Hertz (Hz)	1 Hz
Milliseconds (ms)	10^{-3} s	Kilohertz (kHz)	10^{3} Hz
Microseconds (μs)	10^{-6} s	Megahertz (MHz)	10^{6} Hz
Nanoseconds (ns)	10^{-9} s	Gigahertz (GHz)	10^{9} Hz
Picoseconds (ps)	10^{-12} s	Terahertz (THz)	10^{12} Hz

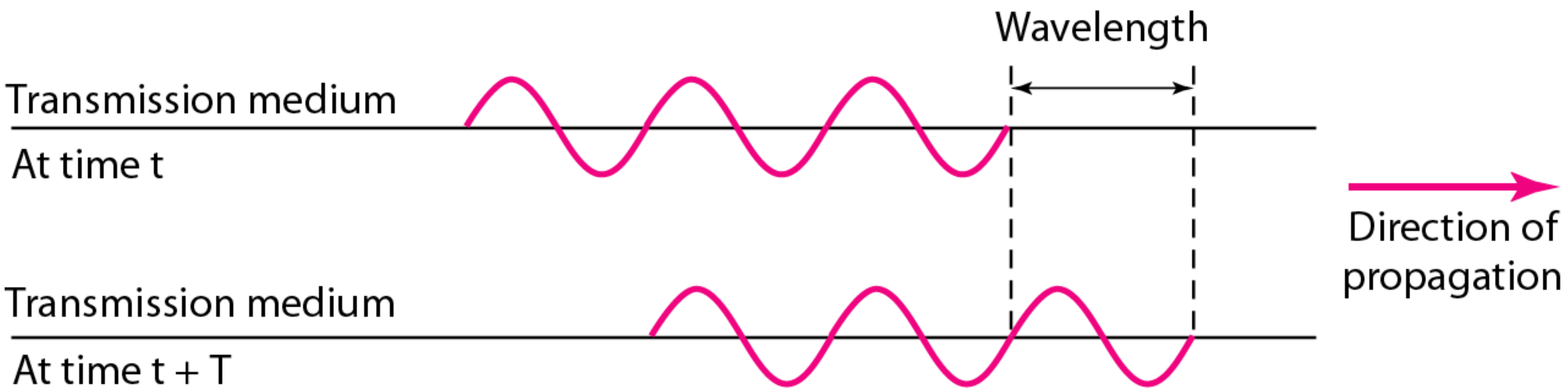
Units of period and frequency

Signal Phase

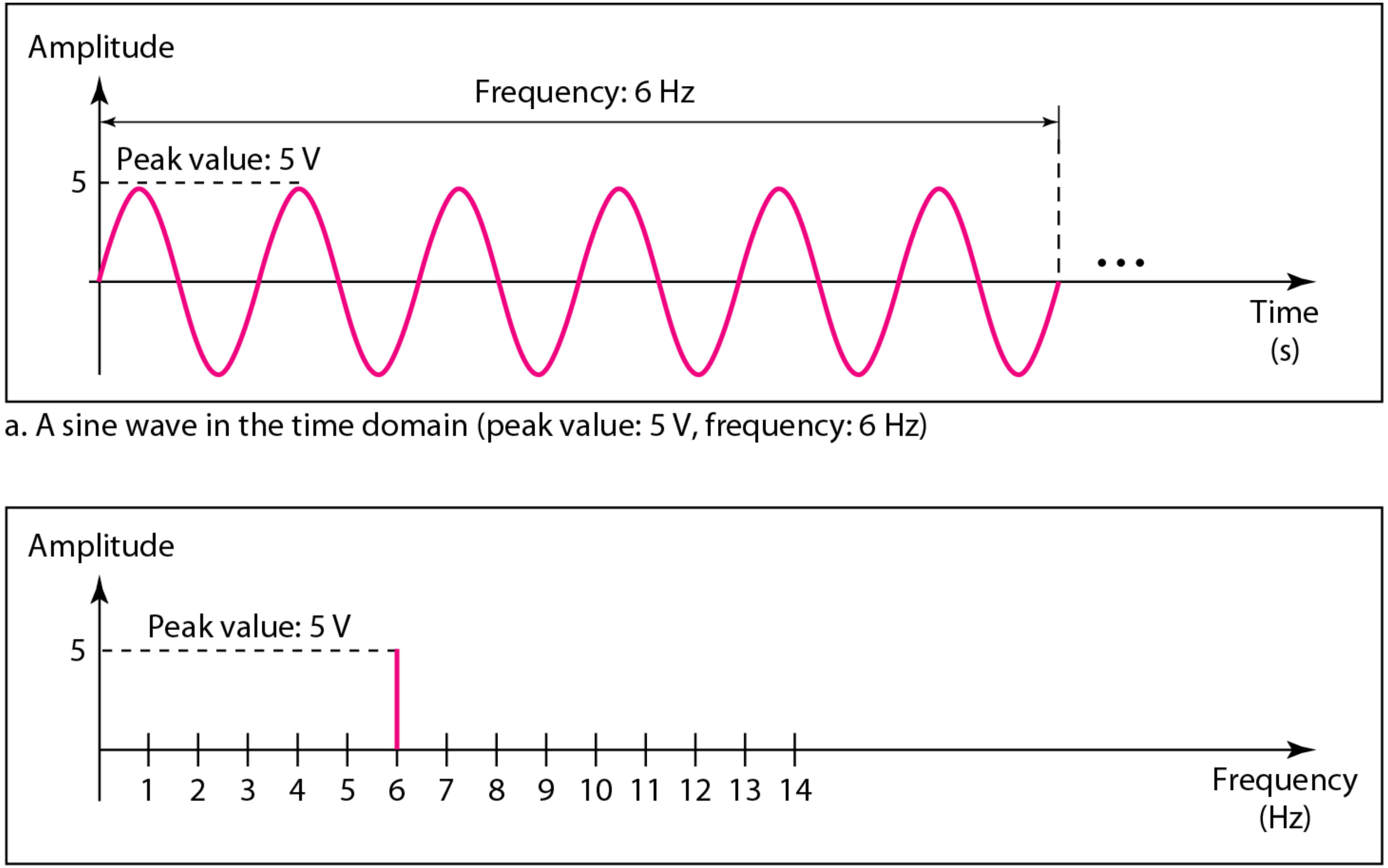


Three signals with different phases

Wavelength and Period



Wavelength and Period

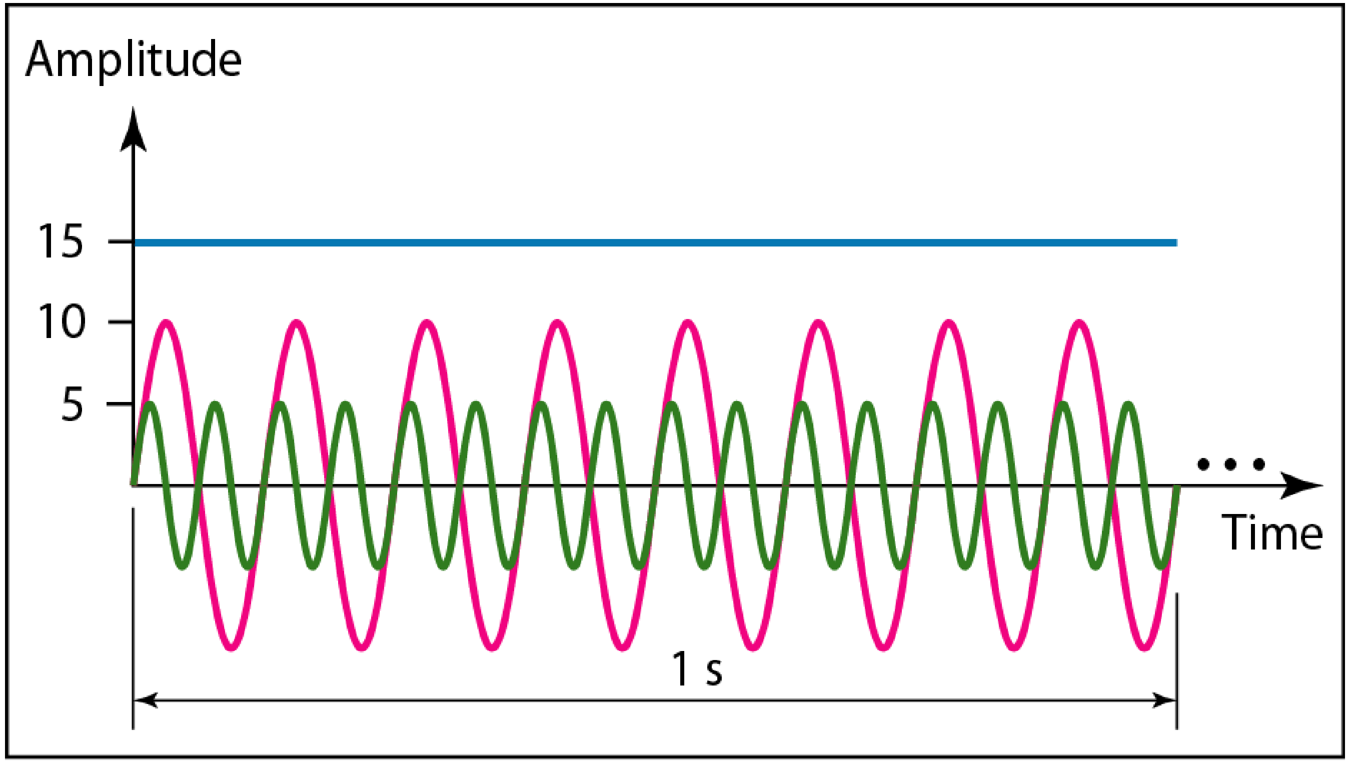


a. A sine wave in the time domain (peak value: 5 V, frequency: 6 Hz)

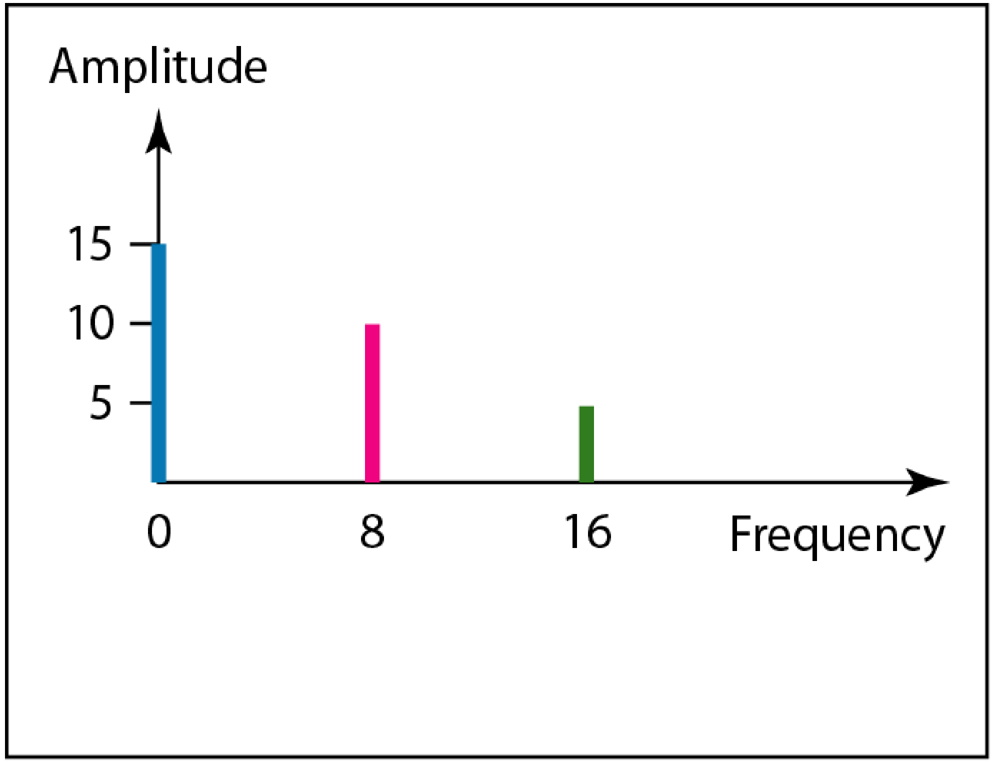
b. The same sine wave in the frequency domain (peak value: 5 V, frequency: 6 Hz)

The time-domain and frequency-domain plots of a sine wave

Wavelength and Period



a. Time-domain representation of three sine waves with frequencies 0, 8, and 16



b. Frequency-domain representation of the same three signals

The time domain and frequency domain of three sine waves

Fourier Analysis ★

- According to Fourier analysis, any composite signal is a combination of simple sine waves with different frequencies, amplitudes, and phases

$$g(t) = \frac{1}{2}c + \sum_{n=1}^{\infty} a_n \sin(2\pi nft) + \sum_{n=1}^{\infty} b_n \cos(2\pi nft)$$

$f = 1/T$: fundamental frequency

a_n and b_n :amplitudes of the n^{th} harmonics

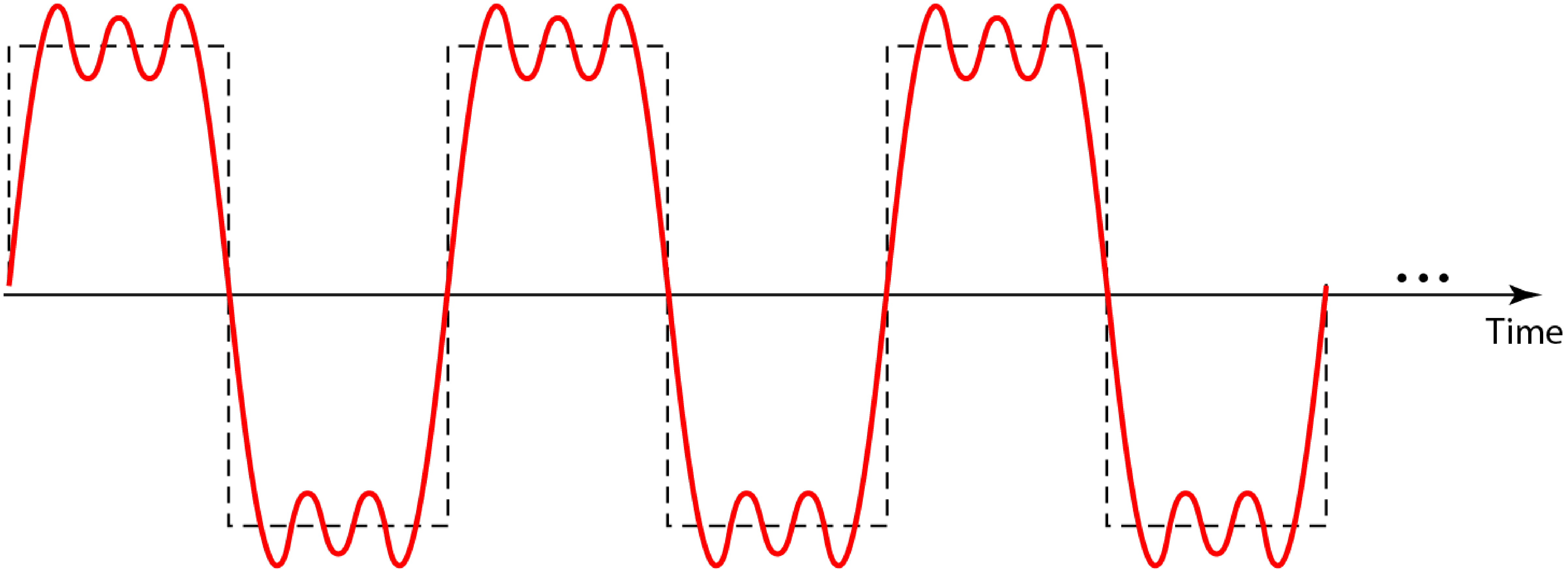
c: a constant.

Signal Decomposition

- If the composite signal is **periodic**, the decomposition gives a series of **signals with discrete frequencies**

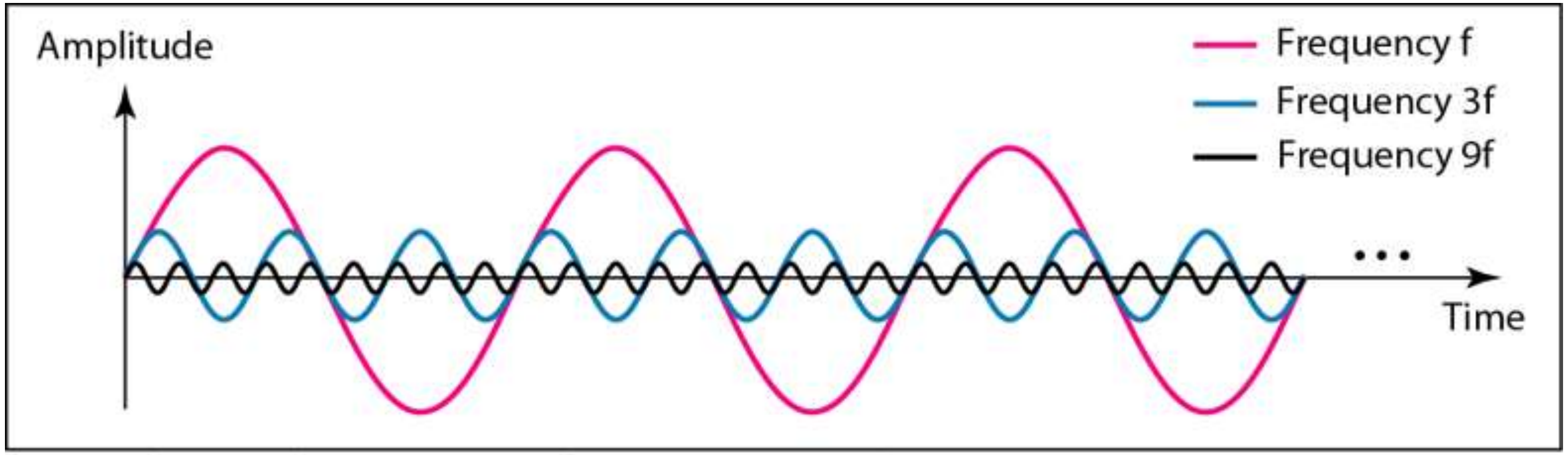
- If the composite signal is **nonperiodic**, the decomposition gives a combination of **sine waves with continuous frequencies**

Signal Decomposition

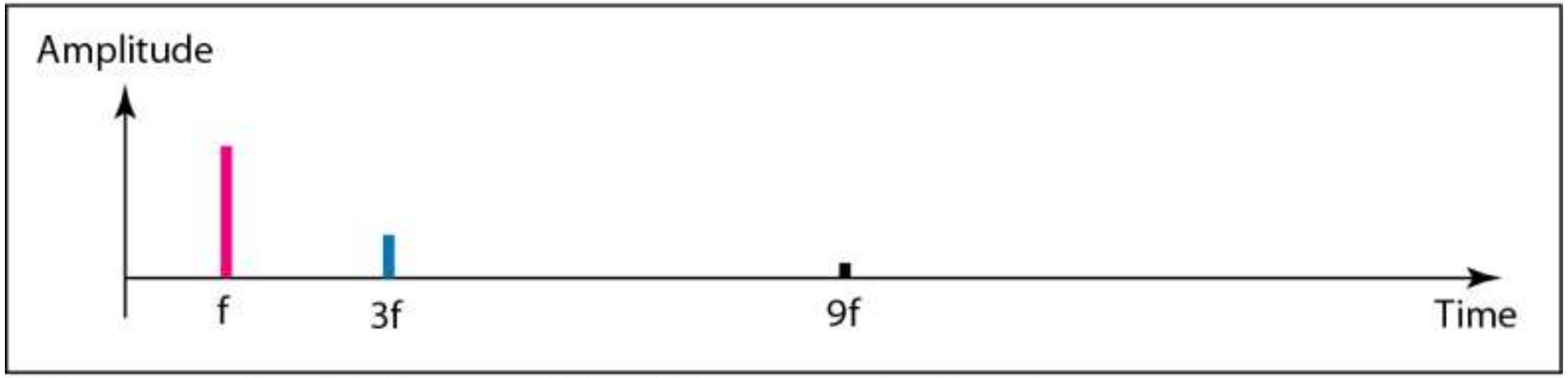


A composite periodic signal

Signal Decomposition



a. Time-domain decomposition of a composite signal

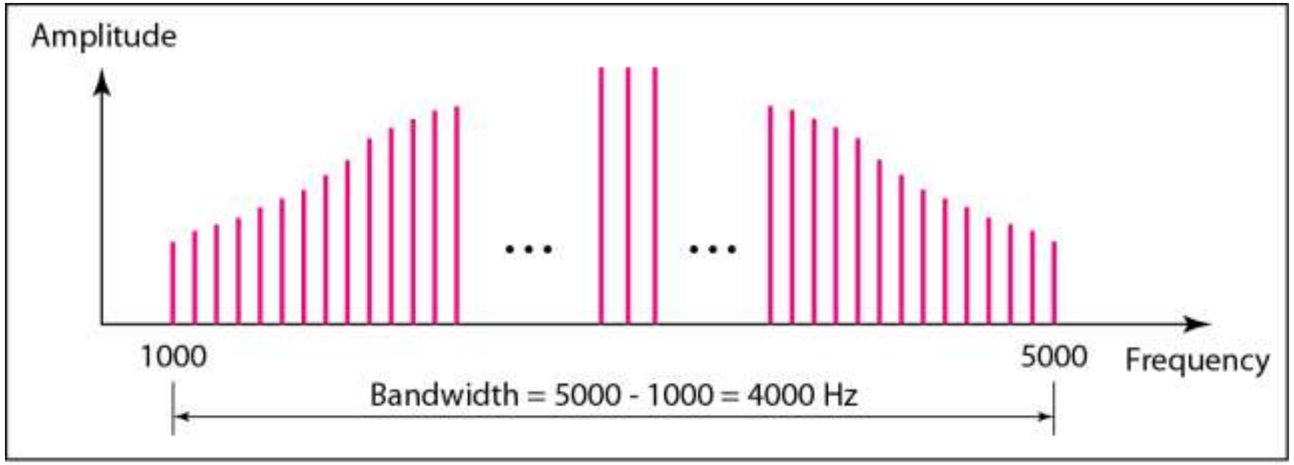


b. Frequency-domain decomposition of the composite signal

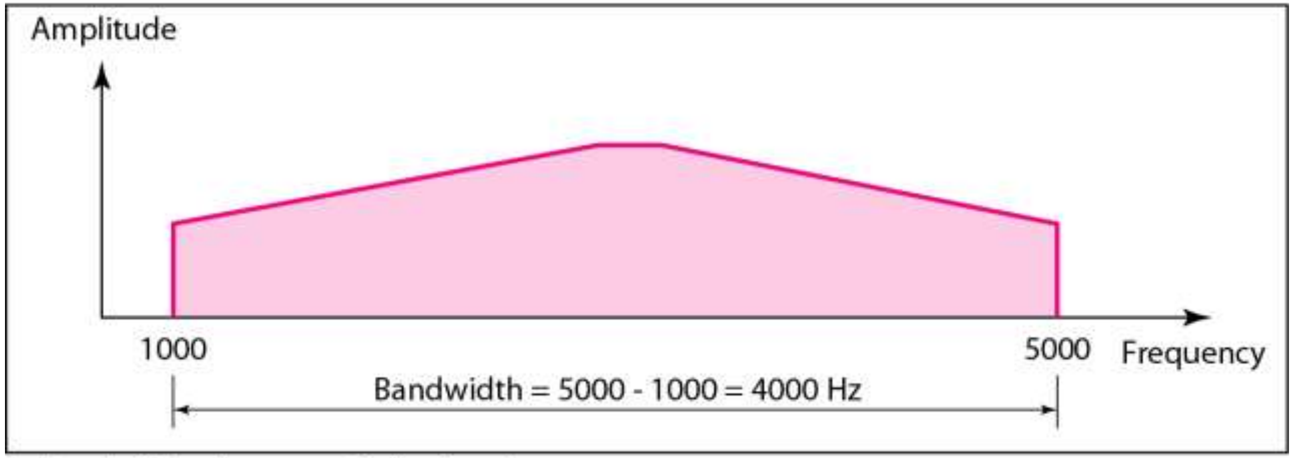
Decomposed signals

Signal Bandwidth

- **Definition:** The bandwidth of a composite signal is the difference between the highest and the lowest frequencies contained in that signal



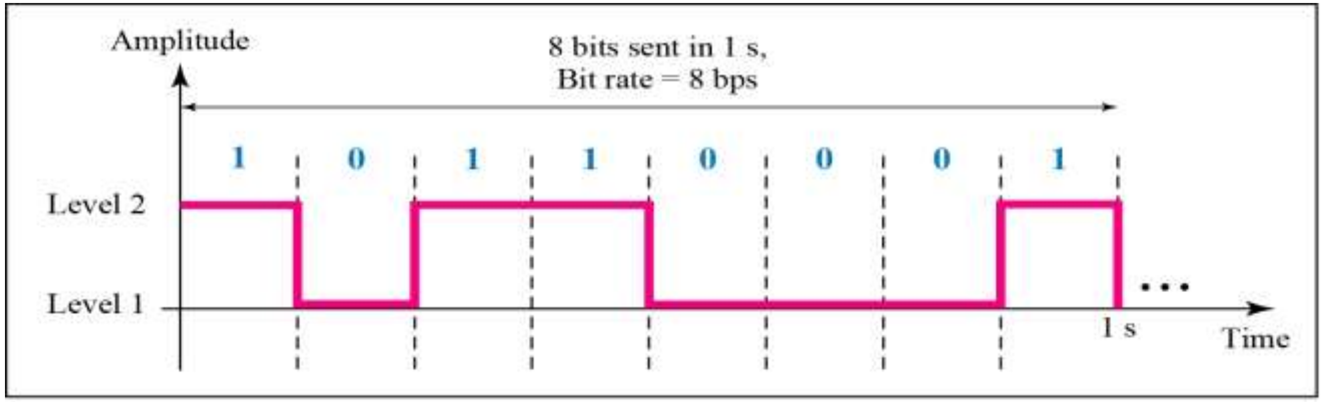
a. Bandwidth of a periodic signal



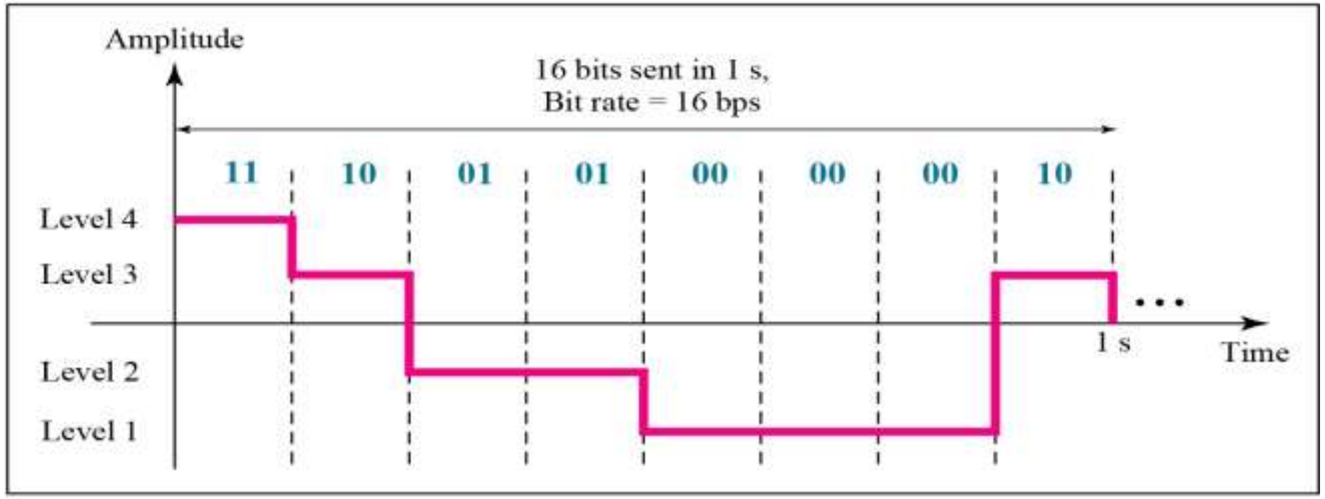
b. Bandwidth of a nonperiodic signal

Digital Signals

- For example, a **1** can be encoded as a positive voltage and a **0** as zero voltage. A digital signal can have **more than two levels**. In this case, we can send more than **1 bit** for each level.

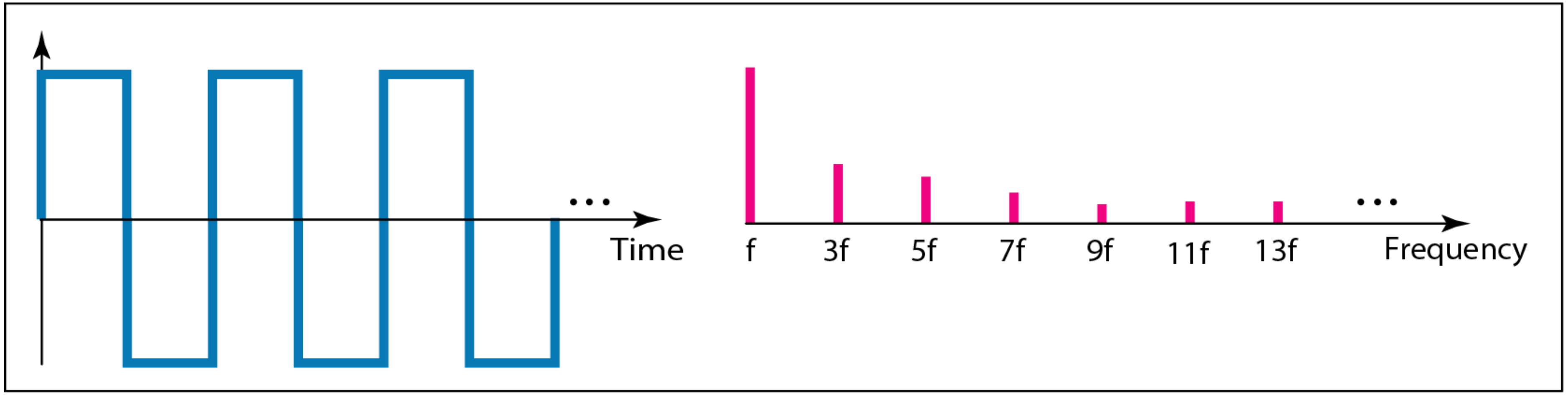


a. A digital signal with two levels

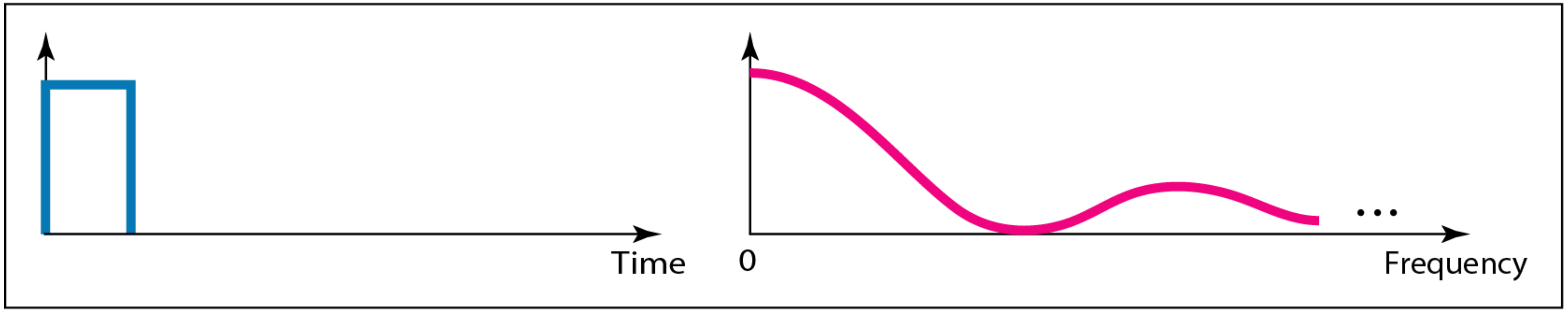


b. A digital signal with four levels

Digital Signals



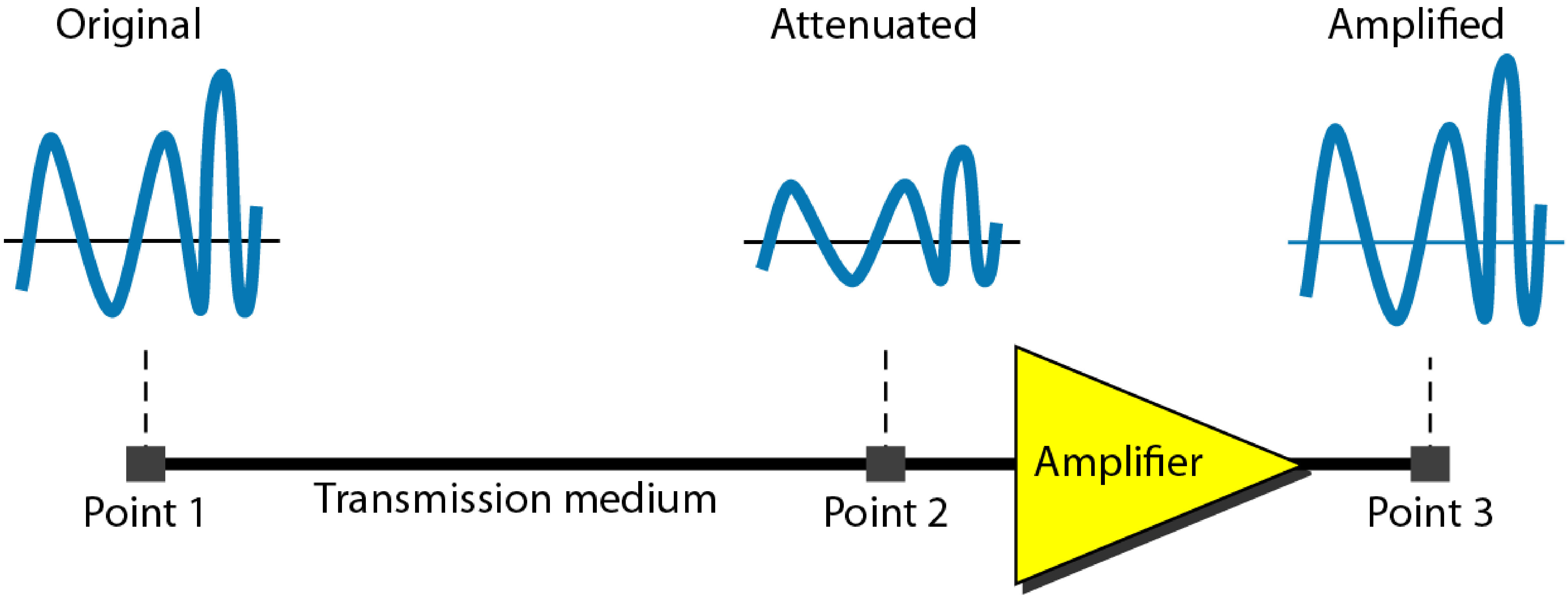
a. Time and frequency domains of periodic digital signal



b. Time and frequency domains of nonperiodic digital signal

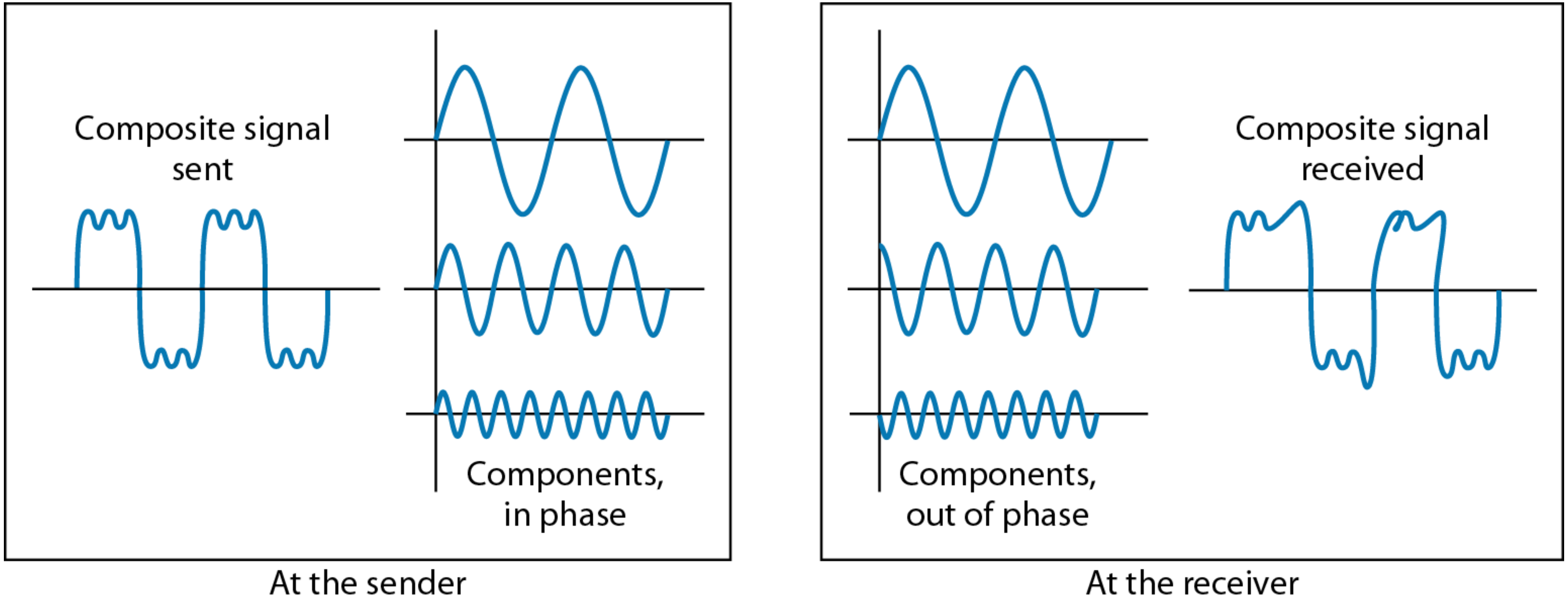
Digital signal can be regarded as a composite analog signal

Transmission Impairment



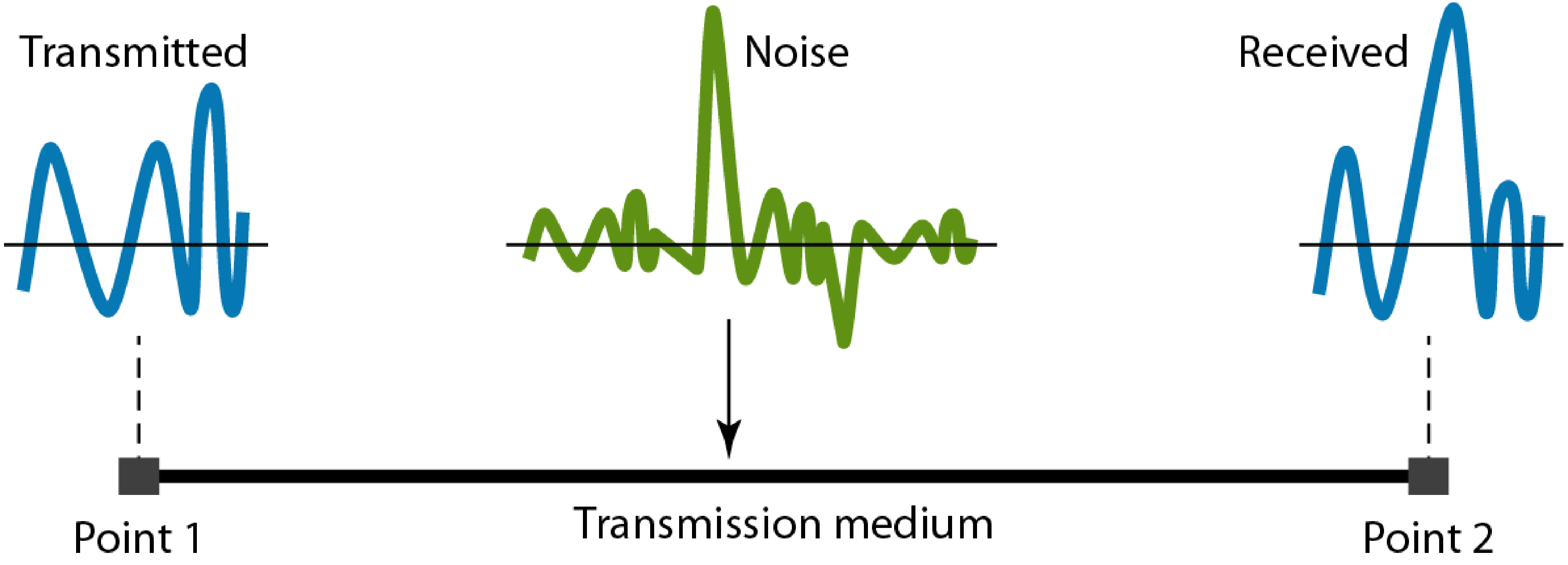
Attenuation

Transmission Impairment



Distortion

Transmission Impairment



Noise

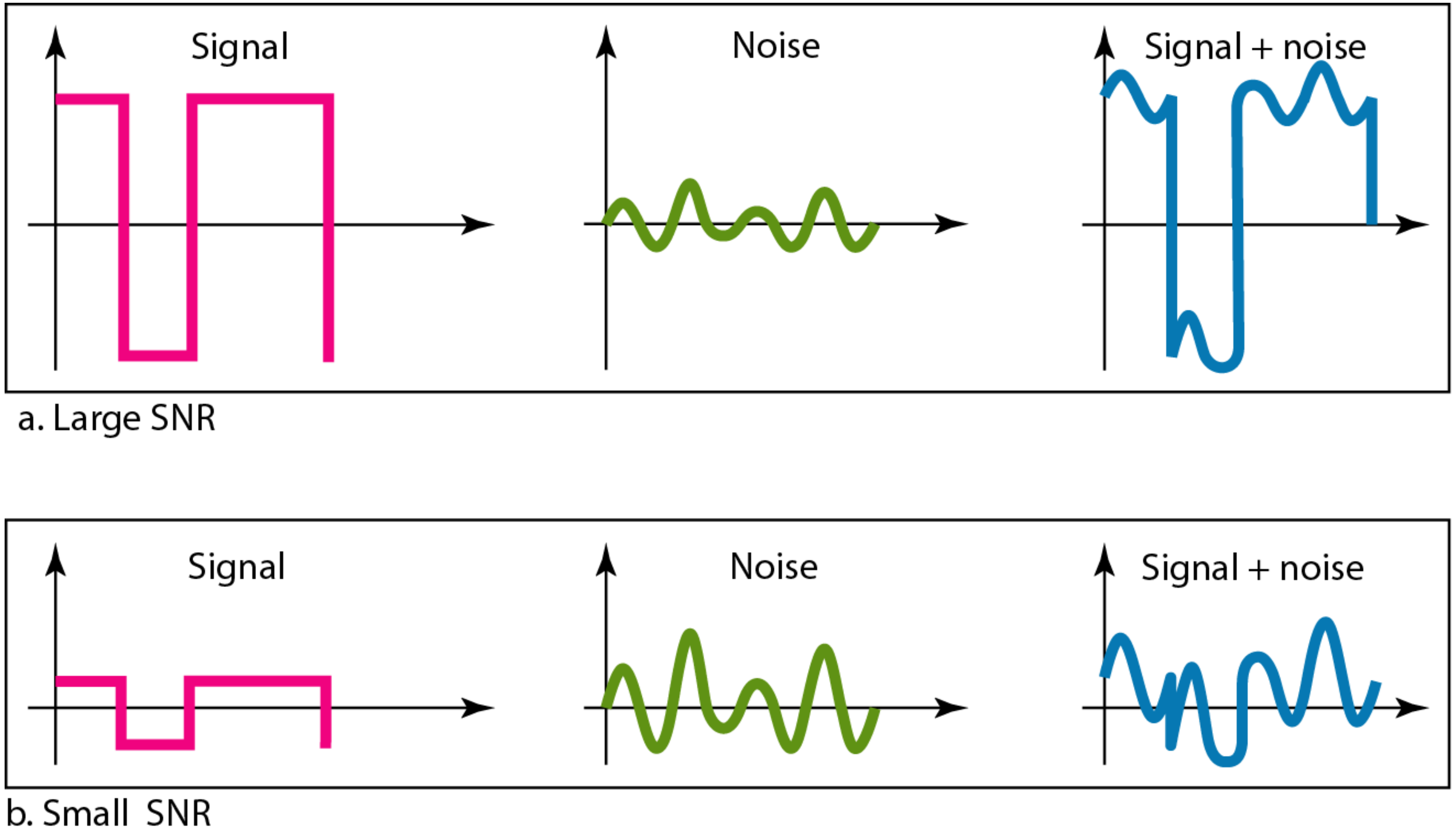
SNR: Signal-to-Noise Ratio

- The power of a signal is 10 mW and the power of the noise is 1 μW; what are the values of SNR and SNR_{dB} ?

$$\mathrm{SNR} = \frac{10{,}000\ \mu\mathrm{W}}{1\ \mu\mathrm{W}} = 10{,}000$$

$$\mathrm{SNR_{dB}} = 10 \log_{10} 10{,}000 = 10 \log_{10} 10^4 = 40$$

SNR: Signal-to-Noise Ratio



Two cases of SNR: a high SNR and a low SNR

Data Rate Limit ★

- Data rate depends on three factors:
  - The bandwidth available
  - The level of the signals we use
  - The quality of the channel (the level of noise)

- Noiseless Channel: **Nyquist Bit Rate**

- Noisy Channel: **Shannon Capacity**

Data Rate Limit ★

Nyquist Bit Rate

- **BitRate = 2 x bandwidth x $\boldsymbol{log_2 L}$**

  L is Number of levels in a signal (two for binary)

Shannon Capacity

- **BitRate = bandwidth x $\boldsymbol{log_2(1 + SNR)}$**

Note: The Shannon capacity gives us the upper limit; the Nyquist formula tells us how many signal levels we need.